Strategic Disclosure of Evidence Produced In the Market

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Abstract

This paper studies how review manipulation affects pricing when evidence about product quality is generated through trade. In a two-period monopoly setting, the initial price impacts both current revenues and the probability a review is produced. The paper analyzes two modes of manipulation, either through censorship of existing bad reviews, or by fabricating a positive one in the absence of trade. The availability of manipulation and associated information distortions change the initial pricing incentives. By adjusting the price, the firm affects the likelihood of trade, which then impacts the equilibrium manipulation strategies and review informativeness. Under full pooling the first-period price may fall or rise compared to no-manipulation benchmark depending on how manipulation changes the marginal informational value of trade. When the price declines it facilitates more trade and raises consumer surplus and total welfare. The mode of manipulation matters with regard to pricing strategies. Censorship and fake reviews result in different price-response thresholds and may induce opposite effects on prices and welfare.

1 Introduction

Typically, the two main factors that determine consumer demand are product's quality and price. When product quality is initially unknown, consumer's willingness to pay reflects that uncertainty. Consumer then relies on the experience of previous buyers to learn about the product. Online reviews provide a natural channel for the transmission of information, however, it can be subject to strategic manipulation. In many settings, firms - regardless of their quality - can manipulate the reviews produced on market by either (i) disclosing only arbitrary portion of reviews, or (ii) adding fake reviews. Ability to suppress unfavorable reviews is a characteristic of social media platforms or company's own websites. On third-party marketplaces and review platforms, manipulation may instead take the form of fabricated reviews.[1] Consumers must therefore interpret the observed information jointly with the possibility that the review process has been manipulated.

The review evidence produced by consumers arises in response to firm's pricing strategy. Consumer can leave an honest feedback only upon purchasing the product, and whether the trade occurs depends on the price set by the firm and the expected quality of the good. Therefore,

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[1] Akesson, Hahn, Metcalfe, and Monti-Nussbaum (2023) study the impact of fake reviews on consumer on a platform resembling Amazon, while Gandhi, Hollenbeck, and Li (2025) conduct an experimental study of fake reviews specifically on Amazon.

the pricing strategy of the firm affects not only current revenues but determines the demand and whether information is produced. This creates a trade-off for the high-quality firm between current revenue and the reputational benefit of generating more information for future consumers. A lower price reduces revenues on the intensive margin, but increases the likelihood of trade and review generation. The initial price influences the expectations of future consumers with respect to the observed evidence, and in turn affects the scope for manipulation and the informativeness of reviews. The consumer's expectations shape the disclosure strategy of both firm types. The initial price is chosen anticipating the resulting immediate revenues and the impact on learning. Additionally, the price may itself carry information about the quality. Such interaction in the model raises the following questions. How does the possibility of review manipulation affect equilibrium prices and demand? When does the manipulation result in an increased welfare? Do these answers depend on the underlying manipulation technology?

These questions are studied using a two-period monopoly model. The firm has a privately known quality and sells its product to a new short-lived consumer in each period. Consumers differ in their privately observed valuations. If the first-period consumer buys the good, she has either a good or a bad experience conditional on the firm's quality, and she generates a truthful review. The firm may conceal the review before it becomes public. The second-period consumer observes the initial price and a public message, but cannot distinguish the absence of first-period trade from the strategic concealment of a generated review. As an alternative manipulation technology considered in section 6, the firm may create a fake review when no genuine review has been generated.

The benchmark setup focuses on a case in which, absent of manipulation, both types pool on the same initial price. Even when there is no direct price signaling, the initial price determines the probability that a review is generated. As the firm of high quality is more likely to obtain a positive review, it has an incentive to price its good below the static monopoly benchmark in order to produce information that may improve its second-period reputation. Availability of the manipulation technology changes this incentive because it affects the informativeness of reviews produced in the initial period. Hence, the pricing strategy depends on how manipulation changes the sensitivity of the high type's expected future reputation to the probability of trade.

The first result shows that the presence of review manipulation can both lower or raise the first-period price. The direction of the effect depends on the impact of manipulation on the informational value of an additional sale. When manipulation increases the marginal informational value of trade, the high type has an incentive to lower its price to generate more evidence. When manipulation decreases this marginal value the incentive to underprice diminishes and the equilibrium price increases relative to the no-manipulation benchmark to boost the initial revenues. A sufficiently large price reduction may also change the form of equilibrium by inducing the low type to separate with some positive probability.

The induced price response links directly to the welfare outcomes. The results show that impact of review manipulation on the consumer need not be determined solely by the associated loss of information accuracy. Under full pooling, manipulation may indeed raise prices, reduce trade, and lower welfare. However, when it strengthens the firm's incentive to generate evidence, it can lead to the initial price decrease and help facilitate more trade in the first period. Under the conditions characterized in the paper, manipulation can create a downward pressure on price and expand the initial trade to generate higher consumer and total surpluses.

In section 6 it is shown that the mode of manipulation matters. Censoring and creating fake reviews alter the informational value of trade in different ways and therefore generate different pricing incentives. Depending on the underlying conditions, both manipulation technologies may lead to a price cut and increase welfare compared to the no-manipulation benchmark.

The model structure is most closely related to two groups of papers. Smirnov and Starkov (2022) study inference under review censorship, while Aköz, Arbatli, and Celik (2020) jointly analyze pricing and biased product reviews. Relative to both settings, the key difference is that review evidence is not exogenously available. The firm's price determines market demand and therefore whether a genuine review is generated. Censorship consequently affects the interpretation of observed messages, as in Smirnov and Starkov (2022), but also the firm's pricing incentives. Unlike Aköz, Arbatli, and Celik (2020), the firm does not directly choose the bias of an exogenous signal, but either manipulates existing evidence or fabricates fake review, where the scope depends on the expected trade likelihood. Hagiu and Wright (2020) and Ispano (2018) instead endogenize the production of information. In the former, purchases today generate information for future consumers but reviews are transmitted directly without any manipulation. In the latter, the sender incurs a separable cost of acquiring evidence and disclosure serves as a signal of self-confidence of the agent. Here, information is produced through revenue-generating trade and, conditional on the available technology, remains subject to strategic review manipulation.

These differences lead to a threefold contribution of this paper. First, the model shows that availability of review manipulation changes the informational return to a marginal increase in sale probability. An increased demand may then limit the scope for strategic disclosure. Depending on these forces the manipulation can therefore either strengthen or weaken underpricing incentives, which in turn feed backs into the initial price. Second, because the response to availability of manipulation operates through price and trade, the paper characterizes the effects on both consumer and total surplus, including conditions under which manipulation increases both. By embedding censorship and fake-review production in the same market environment, the paper also isolates the impact of each manipulation technology on prices and welfare. To the best of my knowledge, this is the first framework to compare these manipulation technologies when the arrival of evidence is endogenous to the firm's pricing decision.

The remainder of the paper is organized as follows. Section 2 discusses the related literature. Section 3 outlines the model and the equilibrium concept used. Section 4 starts by setting up a no-manipulation benchmark before characterizing the effects of manipulation on the pricing strategy and providing sensitivity analysis. Section 5 presents welfare results. The alternative technology of fake reviews is then introduced and compared with censoring in section 6. Finally, section 7 provides a discussion and concludes.

2 Literature

The strand of literature most closely related to this paper connects consumer reviews with strategic pricing or review manipulation. In Smirnov and Starkov (2022), the authors study censorship of unfavorable reviews and show that in the presence of naive and strategic consumers, disclosure of bad review may be an optimal strategy for the good type firm. Aköz, Arbatli, and Celik (2020) instead model manipulation through a signal jamming. Firms choose

both prices and the amount of costly bias introduced through an outside signal. Pricing and reviews also interact in Martin and Shelegia (2021), where reviews depend on the gap between product quality and consumers' prior expectations, resulting in a separating equilibrium as low-type cannot overcharge to avoid bad reviews. Similarly, Aleksenko and Kohlhepp (2025) also model the environment in which price directly affects the value-based feedback. Finally, Wu and Qiu (2023) connect dynamic pricing and fake-review purchases in a competitive market, where consumers may observe a mixture of genuine and fake reviews. The present analysis differs conceptually from the above models by how prices and reviews are intertwined. The price does not itself affect the content of a unbiased, honest review. But it determines the likelihood a review is generated, which then, depending on the underlying technology, influences the scope for manipulation. Such direct connection of price and strategic disclosure produces a feedback effect between manipulation and informational value of trade, and eventually the equilibrium pricing. Then, it enables welfare analysis with respect to the manipulation availability and a comparison between manipulation technologies.

The endogenous arrival of evidence relates this paper to the literature on information production and acquisition. Similarly to Hagiu and Wright (2020), the early consumers generate information about the products which can be used by future customers. However, the authors abstract from any manipulation of the signal, as the firm can only affect the information generation process. Ispano (2018) tackles a problem of a sender who may acquire costly signals whose realization depends on the underlying quality. Then, she decides whether to disclose the information. In that setting, the acquisition of the signal is itself informative regarding the type, and even unfavorable evidence is disclosed. Here, there is no separate evidence acquisition stage. The review generation is a by-product of the interaction with buyers, and then it is subject to manipulation. The initial price affects both the immediate revenues and the likelihood of honest review generation. The evidence produced is not immediately transmitted, but the disclosure is subject to strategic considerations. The information not only increases future learning, but affects the scope for manipulation, which then feed backs into the revenue problem of the firm.

The paper also relates to models of strategic communication on review platforms. Mayzlin (2006) studies promotional messages mixed with word of mouth in an online forum when consumers understand that communication may be strategic, where firms spend more advertising inferior products. Glazer, Herrera, and Perry (2021) show a model in which every posted review may be fake, in which case the learning process slows down. Additionally, any intervention by the platform is counterproductive. Knapp (2026) studies a problem with agents that are either naive or aware of the existence of fake reviews, and shows an equilibrium characterized by a blend of genuine and fake reviews. The above papers focus on the interpretation of manipulated messages conditional on the availability of reviews. In the present model, the firm's pricing strategy first determines the probability of existence of an honest review. The strategic disclosure strategy depends on the availability of the message, and its expected impact affects the initial price choice. The analysis also compares censorship of honest reviews with fabricating of fakes instead of considering each technology in isolation.

The present analysis also connects to a broader literature on the selective disclosure of verifiable evidence. In their seminal work, Milgrom (1981) and Dye (1985) establish, respectively, the mechanism behind an unraveling and the possibility of nondisclosure under the uncertainty with respect to existence of evidence. Gao (2022), Rappoport (2025), and Jiang (2019)

study the inference based on a strategic disclosure when receivers are uncertain about the structure or amount of the evidence possessed by the sender. Here, the model incorporates a simpler evidence space, but although there is at most a single review available, the probability measure allows for a richer set of disclosure strategy. Additionally, the existence of evidence is endogenous to a pricing decision of the firm and affects the interpretation of observed public signal. The pricing decision also connects to Milgrom and Roberts (1986) and Daughety and Reinganum (2008), where market actions and direct claims are combined to signal product quality. Here, the initial price may carry some information, but it plays a crucial role in a pooling equilibrium. Then, by determining the likelihood an honest review was written it affects the inference of consumers from observing a public message. The informational benefit from generating a review creates a downward price force distinct from price signaling.

Finally, the welfare analysis presented connects to work on the regulation of a deceptive product communication. Piccolo, Tedeschi, and Ursino (2015) show that over-deterring of deceptive practices may harm consumers through its effect on competition, while Janssen and Roy (2022) and Rhodes and Wilson (2018) study regulation of product communication and false advertising. These papers indicate that the welfare effects of less biased communication depend on the underlying market interaction, and show that penalizing false claims may be harmful to all types of firm, and optimal policy permits some degree of false advertising. Here, the channel through which strategic communication affects welfare differs. In a monopoly setting, manipulation affects the informational value of a marginal change in trade probability. Under some conditions identified in the paper, the firm has incentives to decrease the price to produce information with higher probability. The welfare consequences come directly from that pricing decision. Increase on the extensive margin allows a high-type firm to curb the negative impact of manipulation, while the increased probability of trade and lower price benefit the consumer and improve market efficiency.

3 Model

We study a two-period market, with time indexed by $t \in \{1, 2\}$, in which a long-lived monopolist sells a product to a new short-lived consumer arriving in each period. The firm's quality is constant over time and privately known, while consumers learn about it from prices and from the review generated through first period trade. The first-period price therefore affects the firm via three channels. In the first period it determines current expected revenues, but can also signal the initial quality (price signaling). The initial price level also affects the probability of sales, therefore whether a review was generated by an early customer. This in turn determines the amount of information available to the second period consumer.

3.1 Firm, consumers, and demand

The monopolist has a privately observed type

$$s \in \mathcal{S} := \{s_L, s_H\}, \qquad 0 < s_L < s_H < 1.$$

The type is drawn once at the beginning of the game and remains fixed across periods. The common initial probability of the high type is

$$\Pr(s = s_H) = \mu_0 \in (0, 1).$$

The firm has zero marginal cost, does not discount future revenue, and cannot commit to its future price or disclosure decision. It maximizes the sum of its expected revenues in the two periods.

A new consumer arrives in each period. A consumer at time t has unit demand and a privately observed valuation

$$v_t \sim G$$

with support $[0, 1]$. Valuations are independently drawn across periods and are independent of the firm's type. The distribution G is common knowledge and admits a density g that is continuous, differentiable, strictly positive on its support, and log-concave.

The realized utility of a consumer with valuation v_t from a purchase at price p_t is given by:

$$u_t(v_t, e_t, p_t) = v_t e_t - p_t,$$

where $e_t \in \{0, 1\}$ denotes the experience from using the good. Conditional on purchasing the product and the product quality s, the probability of a good experience is:

$$\Pr(e_t = 1 \mid s) = s,$$

while the outside option of not buying the product yields zero.

Let $\mu \in [0, 1]$ denote the consumer's posterior belief that the firm is the high type, after observing all public information available at that point. The corresponding expected product quality is:

$$\bar{s}(\mu) = \mu s_H + (1 - \mu) s_L.$$

Before purchasing, the consumer does not observe the random realization of her experience. Hence, given price p and posterior belief μ, her expected utility from purchasing is:

$$\mathbb{E}[u \mid v, p, \mu] = v\bar{s}(\mu) - p.$$

Then, the consumer purchases if and only if the expected utility is non-negative:

$$v\bar{s}(\mu) \geq p,$$

or equivalently,

$$v \geq \frac{p}{\bar{s}(\mu)}.$$

Consequently, for a price p and expected quality $\bar{s}$, the probability of trade is defined as:

$$\lambda(p, \bar{s}) := \Pr\left(v \geq \frac{p}{\bar{s}}\right) = 1 - G\left(\frac{p}{\bar{s}}\right).$$

Throughout the paper, G is understood to take the value zero below its support and one above it, so the expression also covers prices outside the interior demand region.

3.2 Review generation and disclosure

In addition to the information conveyed directly by the first-period price, initial trade may generate review evidence about the firm's quality. Following the first period interaction – and conditional on the consumer purchasing the product – she observes the realization of her experience $e_1 \in \{0, 1\}$ and always leaves an honest review. If no trade takes place, no review is generated. Let λ_1 denote the probability of first-period trade induced by the first-period price, the expected quality of the product and the distribution of consumer's valuation.

Before the firm's disclosure decision, the set of potential reviews is denoted by y, where $y \in \{0, 1, \varnothing\}$, and $y = \varnothing$ occurs only when there was no first-period trade. Conditional on firm quality s and induced probability of trade λ_1, the raw outcome probabilities are:

$$\begin{aligned} \Pr(y = 1 \mid s, p_1) &= \lambda_1 s, \\ \Pr(y = 0 \mid s, p_1) &= \lambda_1 (1 - s), \\ \Pr(y = \varnothing \mid s, p_1) &= 1 - \lambda_1. \end{aligned}$$

The baseline manipulation technology is review censoring. After privately observing a generated review, the firm may choose to either disclose or conceal it. Let

$$\gamma(y \mid s_i, p_1) \in [0, 1]$$

denote the probability with which a firm of type $s_i \in \{s_L, s_H\}$ conceals a review with content $y \in \{0, 1\}$ following the initials sales probability λ_1.

The publicly observed message m belongs to:

$$m \in \mathcal{M} := \{0, 1, \varnothing\}.$$

If a generated review is disclosed, the public message equals its content:

$$m = y.$$

If a generated review is concealed, the public message is:

$$m = \varnothing.$$

No first-period trade also necessarily produces $m = \varnothing$. Hence, the no review message occurs when: i) no trade has taken place in the first period, or ii) the review produced by the initial consumer was concealed.

The second-period consumer observes the first-period price and the public message, but does not observe whether trade took place. Given the observed first-period price and the distribution G, the second-period consumer knows the equilibrium probability of trade and forms beliefs over whether a null message reflects no trade or strategic concealment. The firm cannot fabricate reviews under the baseline censoring technology [2].

[2]Section 6 considers a possibility of producing fake reviews instead of censoring existing ones.

3.3 Strategies and beliefs

We next specify how the firm's pricing and disclosure strategies determine the beliefs of consumers' regarding the quality of the product.

Let

$$\sigma(p \mid s, \mu)$$

denote the likelihood with which a firm of type s posts price p when the current public belief that the firm is the high type is μ. For $t \in \{1, 2\}$, let μ_t denote the posterior belief that the firm is the high type that is relevant for the period-t consumer's purchase decision.

After observing the first-period price, the consumer in $t = 1$ updates the prior belief according to:

$$\mu_1(p_1) = \frac{\mu_0 \, \sigma_1(p_1 \mid s_H, \mu_0)}{\mu_0 \, \sigma_1(p_1 \mid s_H, \mu_0) + (1 - \mu_0) \, \sigma_1(p_1 \mid s_L, \mu_0)}.$$

Before the purchasing decision is made in period $t = 2$, the second-period consumer observes the public message m provided by the firm. Given the review-generation process and the disclosure strategy, the message likelihood is given by $Pr(m|s, p_1, \gamma)$. The second-period consumer updates the belief according to:

$$\mu_2(p_1, m, \gamma) = \frac{\mu_1(p_1) \Pr(m \mid s_H, p_1, \gamma)}{\mu_1(p_1) \Pr(m \mid s_H, p_1, \gamma) + [1 - \mu_1(p_1)] \Pr(m \mid s_L, p_1, \gamma)}.$$

The single-period benchmark in section 4.1 shows that, conditional on μ_2, both firm types choose the same second-period price. Therefore p_2 conveys no additional information, and μ_2 is updated using only the public message regarding the initial consumer's experience.

The expected quality of the firm and the expected probability of trade in either period $t \in \{1, 2\}$ are given by, respectively:

$$\bar{s}_t = \bar{s}\big(\mu_t\big) = \mu_t s_H + (1 - \mu_t) s_L, \qquad \lambda_t = 1 - G\left(\frac{p_t}{\bar{s}_t}\right).$$

On-path beliefs are determined by Bayes' rule. Off-path beliefs and the refinement applied to disclosure decisions are specified in the equilibrium definition below.

3.4 Timing

The timing of the game is depicted in the Figure 1 below. The first-period price determines both current demand and the probability that review evidence is generated, which in turn affects firm's disclosure strategy. Based on the public message second-period consumer updates her belief which determines the willingness-to-pay faced by the firm.

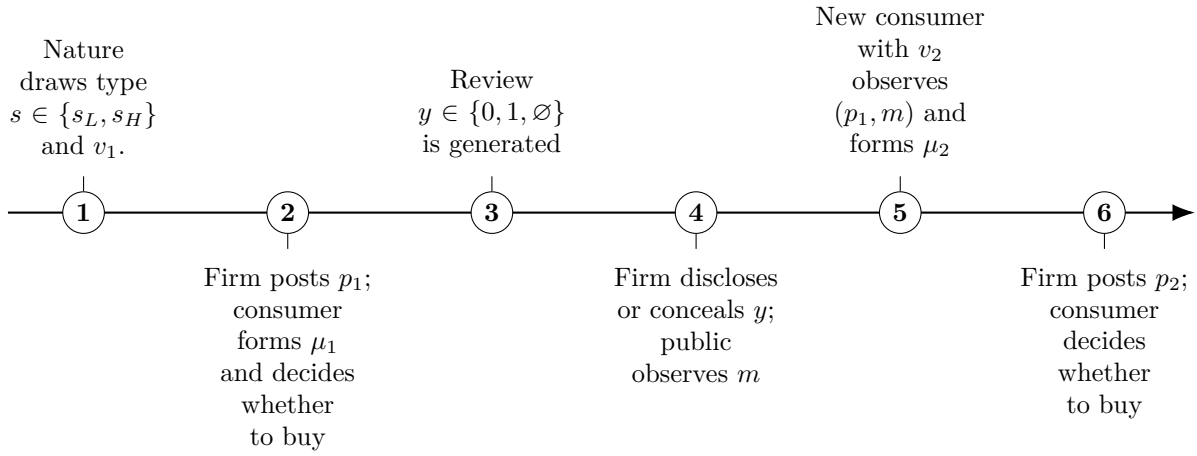


Figure 1: Timing and information flow.

The firm's type and any generated review are privately observed by the firm, whereas prices and the public message are observed by consumers. Consumer valuations are privately observed by the respective consumers. Importantly, following $m = \varnothing$, the second-period consumer cannot distinguish between the absence of first-period trade and strategic concealment of a generated review. This enables the firm to manipulate second-period beliefs through strategic review disclosure.

3.5 Equilibrium concept and benchmark scope

The appropriate equilibrium concept for this game is Perfect Bayesian Equilibrium. An equilibrium specifies the pricing strategies of both firm types, their disclosure strategies, and the public beliefs induced by observed prices and messages. Given these objects, consumers follow the purchase rule defined above, while each firm type maximizes expected revenues over the two periods.

Definition 1 (Perfect Bayesian Equilibrium)**.** *For every firm type $s_i \in S$, define its expected two-period payoff as*

$$\Pi_i = \mathbb{E}\left[R_1(p_1, \bar{s}_1) + R_2(p_2, \bar{s}_2) \mid s = s_i\right],$$

where

$$R_t(p_t, \bar{s}_t) = p_t \lambda(p_t, \bar{s}_t), \qquad \bar{s}_t = \mu_t s_H + (1 - \mu_t) s_L.$$

A Perfect Bayesian Equilibrium is a tuple $\mathcal{E}^* = (\sigma_1^*, \gamma^*, \sigma_2^*, \mu_1^*, \mu_2^*)$ of pricing strategies, review-disclosure probabilities and consistent beliefs such that each type chooses prices and disclosure probabilities to maximize its expected revenues given the overall strategy profile:

$$\left(\sigma_{1,i}^*, \gamma_i^*, \sigma_{2,i}^*\right) \in \arg\max_{\sigma_{1,i}, \gamma_i, \sigma_{2,i}} \Pi_i(\sigma_1, \gamma, \sigma_2, \mu_1^*, \mu_2^*). \qquad i \in \{L, H\}$$

We define the above objects of the equilibrium tuple as follows:

1. *σ_1^* is the 1st period pricing strategy: $\sigma_1^* : S \times [0,1] \to \Delta(\mathbb{R}_+), \quad (s_i, \mu_0) \mapsto \sigma_1^*(\cdot \mid s_i, \mu_0)$.*
2. *γ^* is the disclosure strategy: $\gamma^* : \{0,1\} \times S \times \mathbb{R}_+ \to [0,1], \quad (y, s_i, p_1) \mapsto \gamma^*(y \mid s_i, p_1)$, where $\gamma^*(y \mid s_i, p_1)$ is the probability that type s_i conceals review y given price p_1. Disclosure probabilities determine public message m conditional on realized y.*

3. *σ_2^* is the 2nd period pricing strategy:* $\sigma_2^* : S \times [0,1] \to \Delta(\mathbb{R}_+), \quad (s_i, \mu_2) \mapsto \sigma_2^*(\cdot \mid s_i, \mu_2)$.

4. *μ_1^* and μ_2^* are public belief systems:*

$$\mu_1^* : \mathbb{R}_+ \to [0,1], \qquad p_1 \mapsto \mu_1^*(p_1),$$

and

$$\mu_2^* : \mathbb{R}_+ \times \mathcal{M} \to [0,1], \qquad (p_1, m) \mapsto \mu_2^*(p_1, m).$$

The equilibrium strategies are sequentially rational at every information set. Moreover, consumers are rational and purchase if and only if:

$$v_t \bar{s}_t \geq p_t,$$

and beliefs following every on-path price and public message are consistent with Bayes' rule. Finally, off equilibrium path actions are assumed to result in the worst-possible beliefs[3] for the firm.

When setting p_1, each firm type internalizes how the initial price choice affects current demand, the probability of a review, the subsequent disclosure decision, and the distribution of public beliefs entering period two. The equilibrium characterization below uses this trade-off to determine first-period pricing and the induced form of the equilibrium.

Furthermore, to narrow down the scope of the equilibrium analysis and limit equilibria multiplicity, two additional assumptions are made.

Assumption 1 (Truth-leaning)**.** *Let $\gamma^0 \equiv 0$ denote the truthful-disclosure profile. For every $y \in \{0,1\}$, $s_i \in S$, and p_1, a review is always disclosed if posterior belief $\mu_2(p_1, y, \gamma)$ satisfies:*

$$\mu_2(p_1, \varnothing, \gamma^*) \leq \mu_2(p_1, y, \gamma^0) \implies \gamma^*(y \mid s_i, p_1) = 0.$$

Thus, whenever concealment does not improve the firm's reputation – or increase the belief that it is the good type – the review is disclosed. A review may be concealed with positive probability only if concealment strictly increases the posterior belief that the firm is good relative to truthful disclosure. The assumption eliminates equilibria in which firms hide reviews despite obtaining no reputational benefit from doing so. When multiple disclosure profiles support the same equilibrium posterior beliefs and continuation payoffs, the truth-leaning refinement selects the profile with the highest probability of truthful disclosure.

Finally, the remainder of the paper focuses on the parametrization that results in the case when without the manipulation the pooling equilibrium with respect to period one price is played. This simplifies the analysis, while showcasing the full scope of the price-reviews mechanism.

Assumption 2 (Pooling no-manipulation benchmark)**.** *Scope of the parameters analyzed is such that in the no-manipulation equilibrium, both firm types choose the same first-period pricing strategy:*

$$\sigma_1^{NM}(\cdot \mid s_H, \mu_0) = \sigma_1^{NM}(\cdot \mid s_L, \mu_0).$$

[3]Consumers assign probability one to the low-type:

$$p \notin \bigcup_{s_i \in S} \operatorname{supp} \sigma^*(\cdot \mid s_i, \mu) \quad \implies \quad \mu^*(p) = 0.$$

Assumption 2 focuses on the natural benchmark case in which the low type's pooling gain dominates the cost of mimicking the high type's dynamic price. In this region, absent manipulation, the low type finds it optimal to mimic the high type's dynamic first-period price. This provides a clean benchmark from which to study whether review manipulation changes the price and, possibly, the form of the equilibrium. The assumption is imposed only on the no-manipulation benchmark and does not restrict the equilibrium once manipulation is available. Pooling is natural in this benchmark because the low type benefits from demand based on pooled expected quality rather than on s_L.

4 Equilibrium characterization

This section characterizes equilibrium pricing and disclosure in the two-period game. The initial price selection anticipates the impact on current revenues and demand, which in turn affect the likelihood and informativeness of each review as well as the scope for manipulation. The available signals inform period two consumers and determine the final round revenues. An equilibrium pricing strategy reflects a trade-off between immediate revenues and the information passed on to future consumers.

The analysis proceeds as follows. First, the one-period benchmark is discussed, which determines the equilibrium price and continuation revenue in period 2. Then the two-period game without manipulation is characterized which serves as benchmark when analyzing the impact and scale of the strategic disclosure of information. Finally,the review censoring is introduced and the analysis of how manipulation changes the pricing strategies follows. Throughout the analysis, the focus stays on the case of pooling equilibrium in the no-manipulation benchmark. Manipulation is not restricted to preserve full pooling and may instead induce other strategies.

4.1 The one-period benchmark

Consider the final-period market in which consumer has the posterior expected quality $\bar{s}$. The firm solves:

$$\max_{p\in[0,1]} R^{static}(p,\bar{s}) = p\lambda(p,\bar{s}) = p\left[1-G\left(\frac{p}{\bar{s}}\right)\right].$$

Let

$$r := \frac{p}{\bar{s}}$$

denote the purchase cutoff. The first-order condition is:

$$\frac{\partial\, p\lambda(p,\bar{s})}{\partial\, p} = 0 \quad \Longleftrightarrow \quad 1-G(r)-rg(r)=0.$$

Under the log-concavity assumption, this condition has a unique interior solution $p^{static}(\bar{s})$:

$$p^{static}(\bar{s}) = \bar{s}\,\frac{1-G(r^*)}{g(r^*)} = r^*\bar{s}.$$

Define

$$\kappa := r^*[1-G(r^*)].$$

Optimized one-period revenue is then:

$$R^{\text{static}}(\bar{s}) = \kappa\bar{s}.$$

Lemma 1 (Static game)**.** *Let $\bar{s}$ be an expected quality of the firm. There exists a pooling one-period equilibrium price in which both types charge:*

$$p^{\text{static}}(\bar{s}) = r^* \bar{s}.$$

Moreover, every separating, semi-separating and other pooling candidate equilibrium is payoff-dominated by the above pooling equilibrium.

Proof is left for Appendix A.1.

The second-period of the game is essentially a one-shot interaction since there is no further revenue at stake and the game ends.

Corollary 1 (Second-period price)**.** *The public belief μ_2 implies expected quality:*

$$\bar{s}_2 = \mu_2 s_H + (1 - \mu_2) s_L.$$

In the selected pooling equilibrium of the second-period game, both types charge:

$$p_2^* = p^{\text{static}}(\bar{s}_2) = r^* \bar{s}_2,$$

and obtain continuation revenue:

$$R_2^*(\bar{s}_2) = \kappa \bar{s}_2.$$

In the first-period the $p^{static}(\bar{s})$ is a natural benchmark to use. However, the dynamic nature of the problem – and the fact that price p_1 determines probability of trade and revenue generation – add to the complexity of the firm choice. The remaining analysis studies how this informational effect changes the firm's pricing decision compared to the static benchmark.

4.2 No-manipulation benchmark

Once the solution to the static problem is established, we can consider a two-period problem in the absence of manipulation technology. It provides a benchmark that can be used to determine the effect of manipulation on the equilibrium pricing strategies set by both types, as well as resulting demand, learning and welfare.

First, lets characterize the low-type's outside option when it does not mimic high-type's pricing strategy. Any alternative price immediately reveals its type. Conditional on price separation, subsequent reviews no longer affect low-type's reputation, hence its dynamic pricing problem collapses to two independent static monopoly problems.

Corollary 2. *If the low-type does not mimic pricing strategy of the high type, its optimal first-period price is the solution to a static game:*

$$p^{\text{static}}(s_L),$$

and the total separation payoff is:

$$U_L^{\text{sep}} = 2R^{\text{static}}(s_L) = 2\kappa s_L.$$

Once the low-type firm decides not to pool with the high-type one it reveals own quality to a future buyer. Since the reputational concerns disappear a profit maximizing firm sets a static optimal price $p^{static}(s_L)$ given the belief induced by separation. Lemma 1 indicates the static optimal price which weakly dominates all other separating prices.

In the PBE studied here, the low type's equilibrium pricing strategy can therefore be represented as a mixture of mimicking high-type's pricing strategy and setting an optimal static price. Specifically,

$$p_{1,L} \sim \begin{cases} \sigma_1(\cdot \mid s_H, \mu_0), & \text{with probability } \alpha, \\ p^{\text{static}}(s_L), & \text{with probability } 1-\alpha, \end{cases}$$

where $\alpha \in [0,1]$ denotes the probability with which the low type pools with the high type. Under Assumption 2, the focus of the paper is on the parameterizations that result in a pooling equilibrium in the no-manipulation benchmark, i.e. $\alpha^{NM} = 1$.

To determine when the low type is willing to pool, define the pooling incentive compatibility gap as:

$$\Phi(p_1) := U_L^{\text{pool}}(p_1) - U_L^{\text{sep}}.$$

This implies that

$$\Phi(p_1) \geq 0$$

if and only if the low type weakly prefers pooling at price p_1 to revealing its type and charging the static monopoly price.

Lemma 2 (Pooling cutoff)**.** *Under full pooling, let $\bar{s}_1^{NM} = \mu_0 s_H + (1-\mu_0)s_L$. On the price interval $p_1 \in \left[0, p^{\text{static}}(\bar{s}_1^{NM})\right]$, the pooling IC gap $\Phi(p_1)$ is continuous and strictly increasing. Moreover,*

$$\Phi\big(p^{\text{static}}(\bar{s}_1^{NM})\big) > 0.$$

Then, there exists a unique pooling cutoff:

$$\widehat{p} := \inf\{p_1 : \Phi(p_1) \geq 0\} \in \left[0, p^{\text{static}}(\bar{s}_1^{NM})\right],$$

such that:

$$\Phi(p_1) \geq 0 \quad \Longleftrightarrow \quad p_1 \geq \widehat{p}.$$

Proof is left for Appendix A.3.

Once the initial price is high enough – on the relevant interval – the low-type firm prefers to always pool. As the price goes towards the static optimum the expected revenues increase, while the probability that a negative review is generated decreases as demand goes down. Assumption 2 therefore restricts attention to parameterizations for which the high type's equilibrium pricing strategy without manipulation lies entirely on or above the pooling cutoff. This restriction simplifies equilibrium characterization once the manipulation technology is available.

Then, the high-type's dynamic pricing problem without manipulation can be characterized. Every price in the support of the high type's first-period pricing strategy must solve:

$$\max_{p_1 \in [0,1]} U_H^{NM}(p_1) := R_1\big(p_1, \bar{s}_1^{NM}\big) + \kappa\, \bar{s}_{2,H}^{NM}(\lambda_1(p_1)),$$

where $\bar{s}_{2,H}^{NM}(\lambda_1) := \mathbb{E}\left[\bar{s}_2 \,|\, m(s = s_H, \lambda_1, \gamma \equiv 0), p_1\right]$ is the high type's expected second-period reputation.

Define the no-manipulation learning slope as the shift in the expected quality of a firm in response to a marginal increase in trade λ_1:

$$D_{NM} := \frac{\partial \bar{s}_{2,H}^{NM}(\lambda_1)}{\partial \lambda_1} = \frac{\mu_0(1-\mu_0)^2(s_H - s_L)^3}{\bar{s}_1^{NM}(1-\bar{s}_1^{NM})} > 0.$$

Since D_{NM} is constant,

$$\bar{s}_{2,H}^{NM}(\lambda_1) = \bar{s}_1^{NM} + D_{NM}\lambda_1.$$

To characterize the price choice, define the marginal effect of the first-period price on the high type's expected payoff as:

$$F^{NM}(p_1) := \frac{\partial U_H^{NM}(p_1)}{\partial p_1}.$$

Using the two components of the high-type's payoff:

$$F^{NM}(p_1) = \underbrace{\frac{\partial R_1(p_1, \bar{s}_1^{NM})}{\partial p_1}}_{R_1'(p_1)} + \underbrace{\frac{\partial R_2(\bar{s}_2)}{\partial \bar{s}_2}\frac{\partial \lambda_1(p_1)}{\partial p_1}\frac{\partial \bar{s}_{2,H}^{NM}(\lambda_1)}{\partial \lambda_1}}_{R_2'(p_1)},$$

where: $\frac{\partial R_2(\bar{s}_2)}{\partial \bar{s}_2} = \kappa$ and $\frac{\partial \bar{s}_{2,H}^{NM}(\lambda_1)}{\partial \lambda_1} = D_{NM}$, Then, the marginal effect on high-types' expected payoff can be rewritten as:

$$F^{NM}(p_1) = R_1'(p_1) + \kappa D_{NM}\lambda_{1,p_1}(p_1),$$

where:

$$\frac{\partial R_2}{\partial \bar{s}_2} > 0, \qquad D_{NM} > 0, \qquad \lambda_{1,p_1} < 0 \implies R_2'(p_1) < 0.$$

To obtain an interior solution under no-manipulation, p_1^{NM} has to satisfy FOC:

$$F^{NM}(p_1^{NM}) = R_1'(p_1^{NM}) + \kappa D_{NM}\lambda_{1,p_1}(p_1^{NM}) = 0 \implies R_1'(p_1^{NM}) = -R_2'(p_1^{NM}) > 0.$$

The high-type firm sets such price p_1^{NM} that the $R_1'(p_1^{NM})$ is greater than zero. As $p^{static}(\cdot)$ denotes the optimum of a static game for which $R_1'(p_1^{static}) = 0$, in the dynamic problem the high-type firm sets a price below the static optimum.

Proposition 1 (Optimal Price Without Manipulation)**.** *Under Assumption 2, there is a unique first-period price p_1^{NM} that solves high type's optimal no-manipulation pricing problem and it satisfies:*

$$0 \leq p_1^{NM} < p^{\text{static}}\left(\bar{s}_1^{NM}\right).$$

Proof is left for Appendix A.4.

The high-type sacrifices current-period revenue in order to increase first-period trade and the probability that review evidence is generated. This helps to increase the ex-ante period two reputation for the high-type, as an increase in first-period trade raises the high type's expected second-period reputation at rate D_{NM}. As per assumption 2, the resulting price p_1^{NM} is such that both types pool on it. The set of parameters for which this is true is non-empty[4]

[4]Appendix A.10 shows that such primitive set is non-empty.

4.3 Manipulation strategy

Once the pricing behavior in a dynamic problem is established by Proposition 1, we turn the attention to the setting in which manipulation via censoring is available. Under censoring technology firm can choose an equilibrium probability with which it wants to conceal a review generated by consumer in period one. Depending on the censoring strategy, manipulation alters the frequency with which specific reviews appear as public signal for future buyer. However, manipulation changes also an information content of each signals. E.g. if the low-type firm decides to hide bad review with some positive probability it increases the posterior belief following a negative review, while no review results in a decrease of belief μ_2.

Firms may choose to strategically disclose reviews to raise expected posterior belief $\mu_2(m)$ and ensure higher revenues in period two. As high-type firm has higher likelihood of producing a good review compared to the low-type firm, absent of manipulation the reviews would have the following ordering with respect to induced posterior beliefs:

$$\mu_2(m = 1) > \mu_2(m = \emptyset) > \mu_2(m = 0).$$

Then, the natural direction of manipulation is to censor a bad review with some positive probability. However, in order for the concealment strategy to constitute part of an equilibrium, it must not create any profitable deviation. Therefore, censorship must be limited in scope, otherwise it could create a situation in which:

$$\mu_2(m = \emptyset) < \mu_2(m = 0).$$

Lemma 3 (Concealment strategy)**.** *In any truth-leaning pooling and partial-pooling PBE, concealment strategy is such that public message m satisfies $\mu_2(m = 0) = \mu_2(m = \emptyset)$. Hiding good reviews is payoff-dominated. Moreover, only the low type hides reviews with positive probability $\gamma_H = 0$, $\gamma_L \geq 0$, where:*

$$\gamma_L^*(\lambda_1) = \frac{(1 - \lambda_1)(s_H - s_L)}{(1 - s_L)\,(1 - \lambda_1 s_H)}, \qquad \lambda_1 = 1 - G\big(p_1/\bar{s}_1\big).$$

Proof is left for Appendix A.5.

The probability of hiding a bad review is picked to ensure that the posterior beliefs following messages $m = 0$ and $m = \emptyset$ are equal. Otherwise, a firm would have an incentive to either hide more or less than the equilibrium strategy prescribes. As good review increases the posterior belief compared to the prior, it increases the price that a firm can charge in period two. Hence, a unilateral censoring of good reviews by any type is dominated by revealing every good review. Moreover, the high-type firm never participates in censoring, as by hiding bad review with a positive probability it would allow the low-type to expand the censoring. If $\gamma_H > 0$ then it allows for γ_L to go up in equilibrium. As such censoring does not help the high-type, by Assumption 1 it reveals reviews truthfully.

Corollary 3. *Equilibrium concealment probability $\gamma_L^* \in [0, 1]$ is monotonically decreasing in λ_1, and in particular $\gamma_L^*(1) = 0$. Furthermore, merging $\{\emptyset, 0\}$ shifts beliefs for "no positive evidence" affecting learning under manipulation.*

As showed in Lemma 3, the concealment probability is a function of early demand λ_1, it reacts to initial price p_1. Lowering p_1 raises λ_1, which reduces γ_L^* and partially restores informativeness. The informativeness of a marginal increase in trade is also affected by the possibility of manipulation.

4.4 Manipulation equilibrium characterization

Establishing the manipulation strategies employed by both types and their properties allows to characterize the equilibrium in the case when manipulation is available. High-type firm reports the reviews truthfully, however, the low-type can benefit from a possibility to censor bad reviews with positive probability. Manipulation probability enters the dynamic pricing problem and affects it through two main channels: (i) price-review channel at a fixed pooling probability, and (ii) pooling feedback. The first channel is always active – switching from no-manipulation to manipulation changes the expected posterior belief as the public message may be strategically altered. This in turn may result in a different pricing strategy choice by the high-type firm. As per Corollary 3, the scope for concealment decreases in the initial demand, hence adjusting the first-period price can be used to limit effective review censoring. The latter channel activates when the initial price change between the manipulation and no-manipulation regimes results in some degree of price separation in period one. Then, due to separation the expected quality in period one, total trade at any given price and belief updating at every message m changes compared to the pooling case.

Substituting the equilibrium concealment strategy $\gamma_L^*(\lambda_1)$, let $U_H^M(p_1, \alpha)$ denote the high type's reduced expected payoff under manipulation. Along any differentiable equilibrium branch $\alpha = \alpha^M(p_1)$, the marginal effect of the first-period price can be decomposed as

$$\frac{dU_H^M\big(p_1, \alpha^M(p_1)\big)}{dp_1} = \underbrace{\frac{\partial U_H^M(p_1,\alpha)}{\partial p_1}\bigg|_{\alpha=\alpha^M(p_1)}}_{\text{price-review channel at fixed } \alpha} + \underbrace{\frac{\partial U_H^M(p_1,\alpha)}{\partial \alpha}\bigg|_{\alpha=\alpha^M(p_1)} \alpha^{M\prime}(p_1)}_{\text{pooling feedback}}. \tag{1}$$

The first component includes the direct effect of the price on current revenue and its indirect effect on review arrival and equilibrium concealment, holding the pooling probability fixed. The second component captures the change in current demand and continuation beliefs generated by the low type's endogenous pooling response.

The analysis focuses first on the case of full-pooling, $\alpha^M = 1$, in which the pooling feedback channel is inactive. The effect of manipulation on the price can then be isolated by comparing the value of review generation under manipulation and no manipulation. If the candidate price remains above the pooling cutoff, it constitutes the equilibrium price. If the price falls below the cutoff it cannot be sustained as a pooling PBE. Then, α^M becomes endogenous, and the pooling-feedback channel must also be taken into account. Finally, the equilibrium is characterized.

4.4.1 The pooling price

Before comparing the pricing incentives across regimes, it is worth noting that the no-manipulation price remains within the full-pooling region when censoring becomes available. Let $\Phi^{NM}(p_1)$ and $\widehat{p}^{NM}$ denote the pooling IC gap and cutoff characterized in Section 4.2, and let $\Phi^M(p_1, 1)$ and $\widehat{p}^M$ denote their counterparts under manipulation when the low type pools with probability one.

The equilibrium concealment strategy weakly increases the low type's continuation payoff from pooling. Hence, on the relevant price interval,

$$\Phi^M(p_1, 1) \geq \Phi^{NM}(p_1),$$

which implies

$$\widehat{p}^{M} \leq \widehat{p}^{NM}.$$

Thus, the benchmark no-manipulation price remains compatible with full pooling after manipulation is introduced. This allows the manipulated and no-manipulation first-order conditions to be compared at p_1^{NM}, while holding $\alpha = 1$.

Let $F^M(p_1)$ denote the marginal effect of the first-period price on the high type's reduced payoff on the pooling branch: $F^M(p_1) := \frac{\partial U_H^M(p_1,\alpha)}{\partial p_1}\Big|_{\alpha=1}$.

Using the two components of the high type's payoff, this marginal effect can be written as

$$F^M(p_1) = \underbrace{\frac{\partial R_1\left(p_1, \bar{s}_1^{NM}\right)}{\partial p_1}}_{R_1'(p_1)} + \underbrace{\frac{\partial R_2\left(\bar{s}_{2,H}^{M}\right)}{\partial \bar{s}_{2,H}^{M}} \frac{\partial \lambda_1(p_1)}{\partial p_1} \frac{d\bar{s}_{2,H}^{M}(\lambda_1)}{d\lambda_1}}_{R_2^{M\prime}(p_1)},$$

where $\bar{s}_{2,H}^{M}(\lambda_1) := \mathbb{E}\left[\bar{s}_2 \mid s = s_H,\, \lambda_1,\, \gamma_L^*(\lambda_1),\, \alpha = 1\right]$ denotes the high type's expected second-period reputation under manipulation on the pooling branch. Compared to the no-manipulation benchmark, here the second-period reputation of the high-type firm depends on the effectiveness of manipulation which enters via $\gamma_L^*(\lambda_1)$ and is affected by the initial pricing strategy.

Define the manipulation learning slope:

$$D_M(\lambda_1) := \frac{d\bar{s}_{2,H}^{M}(\lambda_1)}{d\lambda_1}.$$

Because the equilibrium concealment strategy $\gamma_L^*(\lambda_1)$ has already been substituted into $\bar{s}_{2,H}^{M}$, $D_M(\lambda_1)$ captures both the direct effect of additional review arrival and the induced response of equilibrium concealment. At fixed $\alpha = 1$, current revenue and the sensitivity of trade to price are unchanged relative to the no-manipulation benchmark; manipulation affects the pricing problem solely through the learning slope $D_M(\lambda_1)$.

Since the optimized second-period revenue satisfies $\frac{\partial R_2(\bar{s}_2)}{\partial \bar{s}_2} = \kappa$, the marginal continuation effect can be rewritten as:

$$R_2^{M\prime}(p_1) = \kappa D_M(\lambda_1(p_1))\, \lambda_{1,p_1}(p_1).$$

Substituting that expression into the marginal effect of a price change in period one results in:

$$F^M(p_1) = R_1'(p_1) + \kappa D_M(\lambda_1(p_1))\, \lambda_{1,p_1}(p_1).$$

To isolate the price force generated by manipulation, evaluate the two regimes at the no-manipulation equilibrium price p_1^{NM}, which was set to result in $F^{NM}(p_1^{NM}) = 0$ for any optimal pooling no-manipulation price $p_1^{NM} \in (0, p^{static}(\bar{s}_1))$[5].

[5]In a boundary case when $p_1^{NM} = 0$, the resulting trade probability is $\lambda_1^{NM} = 1$. Lemma 3 implies that $\gamma_L^*(\lambda_1^{NM}) = 0$. As in that case there is no manipulation, for a full-pooling price $p_1 = 0$:

$$\bar{s}_{2,H}^{M}(1) = \bar{s}_{2,H}^{NM}(1).$$

Since at $p_1^{NM} = 0$ manipulation does not affect the problem, the optimal pooling price remains unchanged, $p_1^{M} = 0$. Assumption 2 implies $\widehat{p}^{NM} = 0$, while $\widehat{p}^{M} \leq \widehat{p}^{NM}$ results in $\widehat{p}^{M} = 0$. The boundary case is formalized in the proof of Proposition 3. The remainder of the subsection focuses on the non-degenerate case $p_1^{NM} > 0$, for which $F^{NM}(p_1^{NM}) = 0$.

Define the shift in the high type's marginal pricing incentive as:

$$\Delta F := F^{M}(p_1^{NM}) - F^{NM}(p_1^{NM}).$$

When $\alpha = 1$, the first-period expected quality, demand, and current-revenue function are the same in both regimes and both terms cancel out. Then, the direction of a price change depends solely on the period two effects:

$$\Delta F = R_2^{M\prime}(p_1^{NM}) - R_2^{NM\prime}(p_1^{NM}).$$

Since $F^{NM}(p_1^{NM}) = 0$, ΔF describes the marginal payoff effect of increasing the price under manipulation, evaluated at the no-manipulation price:

$$F^{M}(p_1^{NM}) = \Delta F.$$

Hence, $\Delta F < 0$ represents a local force toward a lower first-period price, whereas $\Delta F > 0$ represents a local force toward a higher price.

In order to use this local force to determine a global shift of high-type's preferred pooling price, the attention is restricted to the case in which the high-type's payoff under the pooling is single-peaked in p_1. The corresponding sufficient regularity condition and its implication is derived in Appendix A.8. Under this regularity condition the high type's preferred full-pooling is unique, and the sign of ΔF determines on which side of p_1^{NM} it lies.

Breaking down the marginal effects of a price change on the period two revenues and substituting in the learning slopes gives:

$$\Delta F = \frac{\partial R_2(\bar{s}_2)}{\partial \bar{s}_2}\lambda_{1,p_1}(p_1^{NM})\left[D_M(\lambda_1(p_1^{NM})) - D_{NM}\right] = \kappa\lambda_{1,p_1}(p_1^{NM})\underbrace{\left[D_M(\lambda_1(p_1^{NM})) - D_{NM}\right]}_{\Delta \text{ in learning slopes}}.$$

As $\kappa > 0$ and $\lambda_{1,p_1}(p_1^{NM}) < 0$ it follows that:

$$\operatorname{sign}(\Delta F) = -\operatorname{sign}\left[D_M(\lambda_1^{NM}) - D_{NM}\right].$$

Under full pooling, the learning slopes are:

$$D_M(\lambda_1^{NM}) = \frac{\mu_0(1-\mu_0)^2(s_H - s_L)^3}{\bar{s}_1^{NM}\left(1 - \lambda_1^{NM}\bar{s}_1^{NM}\right)^2}, \quad \text{and} \quad D_{NM} = \frac{\mu_0(1-\mu_0)^2(s_H - s_L)^3}{\bar{s}_1^{NM}\left(1 - \bar{s}_1^{NM}\right)}.$$

Comparison of these two expressions yields:

$$D_M(\lambda_1^{NM}) > D_{NM} \quad \Longleftrightarrow \quad \lambda_1^{NM} > \lambda^* := \frac{1}{1 + \sqrt{1 - \bar{s}_1^{NM}}}.$$

Therefore,

$$\lambda_1^{NM} \gtrless \lambda^* \quad \Longleftrightarrow \quad D_M(\lambda_1^{NM}) \gtrless D_{NM} \quad \Longleftrightarrow \quad \Delta F \lesseqgtr 0.$$

Proposition 2 (Pooling price response)**.** *Suppose $p_1^{NM} > 0$ and the full-pooling regularity condition holds. Let $\Delta F \equiv F^M(p_1^{NM}) - F^{NM}(p_1^{NM})$ be the shift in the high type FOC at the no–manipulation price. If the price induced by manipulation remains above the pooling cutoff,*

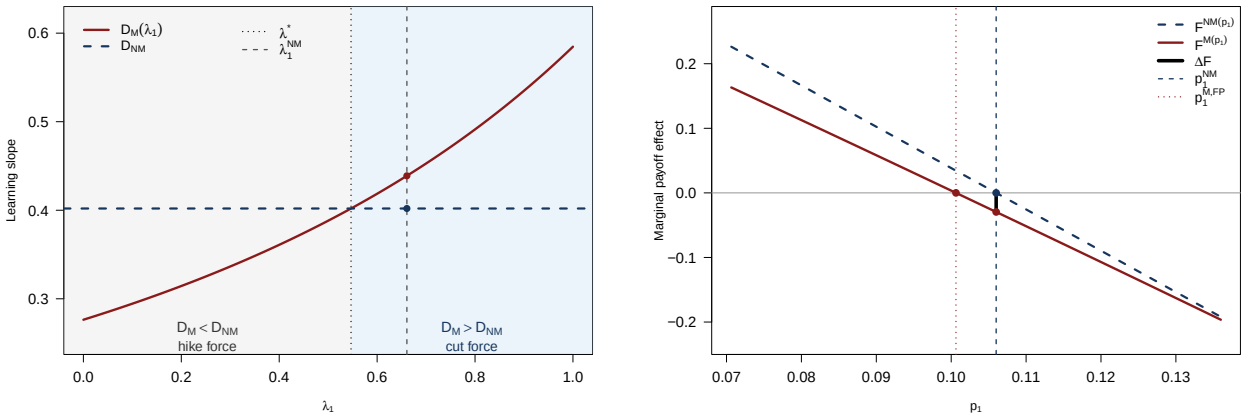
$$p_1^M \geq \widehat{p}^M,$$

the pooling equilibrium price satisfies:

$$\lambda_1^{NM} > \lambda^* \implies p_1^M < p_1^{NM}, \quad \lambda_1^{NM} < \lambda^* \implies p_1^M > p_1^{NM}.$$

Proof is left for Appendix A.8.

If $p_1^M \geq \widehat{p}^M$ holds, the low type's pooling incentive constraint is satisfied and p_1^M is supported as pooling equilibrium price. When $\lambda_1^{NM} < \lambda^*$, no trade is sufficiently frequent that concealed bad reviews can be pooled with a common empty message. Censorship then flattens learning, lowers the informational value of an additional sale, and creates a force toward a higher price. When $\lambda_1^{NM} > \lambda^*$, review is relatively likely to be generated and reputation gain from concealment decreases rapidly with the probability of trade. The empty message becomes increasingly unfavorable as demand goes up, which steepens learning and creates a downward force on the first-period price.



(a) Learning slopes

(b) Shift in the pricing first-order condition

Figure 2: Pooling price response to review censoring at: $G = U[0,1]$, $\mu_0 = 0.25$, $s_L = 0.10$, $s_H = 0.95$. Panel (a) compares learning slopes $D_M(\lambda_1)$ and D_{NM}. The intersection at λ^* separates the price-increase and price-reduction regions. Panel (b) illustrates $\Delta F < 0$ resulting in a downward shift of the pooling price.

The example in Figure 2 illustrates the mechanism. In Panel (a), the demand level satisfies $\lambda_1^{NM} > \lambda^*$ and the manipulation learning slope is above the benchmark one. This pushes the first-order condition with respect to the second-period down. The initial equilibrium price has to go down compared to the benchmark for the FOC to be satisfied. This example highlights that the sign of the price effect is governed by the marginal informational value of additional trade. Appendix B.1 contains an example of a specification for which the high-type firm hikes the initial price in response to manipulation.

A manipulation-induced price increase necessarily preserves full pooling, while a price reduction does not result in any degree of period one separation as long as

$$p_1^M \geq \widehat{p}^M.$$

The next subsection focuses on the situation in which the pooling incentive constrain of the low-type firm does not hold, and its pooling probability adjusts endogenously.

4.4.2 The semi-separating price

The remaining case arises when the manipulation-induced price falls below the pooling cutoff:

$$p_1^M < \widehat{p}^M.$$

At this price, always mimicking the high-type is no longer optimal for the low-type, and the probability of pooling drops below one. The low type instead pools with some probability $\alpha^M \in (0,1)$, which is determined by the first-period price and the condition $\Phi^M(p_1, \alpha) = 0$. Then, the firm composition on the pooling branch responds to the price affecting the expected quality, demand, and continuation beliefs. This in turn activates the pooling-feedback channel introduced in equation (1).

The review channel continues to determine the initial direction of the price change relative to the no-manipulation level. If the price goes up towards the optimal static level the equilibrium remains pooling. However, once the price declines it may relax the incentive condition of the low-type to mimic.

Conditional on the price decrease below the full-pooling cutoff $\hat{p}^M$ and α^M drops, the pooling feedback channel starts to affect the revenues and impacts the equilibrium price choice. After observing the pooling price, the posterior probability and the corresponding expected quality of the firm on the pooling branch become:

$$\mu_1(\alpha) = \frac{\mu_0}{\mu_0 + (1-\mu_0)\alpha} \quad \Longrightarrow \quad \bar{s}_1(\alpha) = \frac{\mu_0 s_H + (1-\mu_0)\alpha s_L}{\mu_0 + (1-\mu_0)\alpha}.$$

The resulting expected quality is strictly decreasing in α. Then, the probability of first-period trade becomes a function of the pooling probability. At an unchanged price it is strictly increasing in the expected quality and decreasing in the pooling probability[6].

Similarly to the no-manipulation case, the low type's pooling incentive compatibility gap under manipulation is given by:

$$\Phi^M(p_1, \alpha) := U_L^{\text{pool},M}(p_1, \alpha) - U_L^{\text{sep}},$$

where a semi-separating equilibrium requires $\Phi^M(p_1, \alpha) = 0$ and $\alpha \in (0,1)$. Additionally, at the pooling cutoff price $\Phi^M(\widehat{p}^M, 1) = 0$ holds. For a given price the incentive constraint may potentially admit more than one value of α. On the relevant price region and for the fixed value of α it holds that:

$$\Phi_{p_1}^M(p_1, \alpha) > 0.$$

This implies that, for a fixed pooling probability α, the low type's incentive constraint determines at most one locally compatible pooling price. Since $\Phi_{p_1}^M(\widehat{p}^M, 1) > 0$, in a one-sided neighborhood of the full-pooling cutoff the implicit-function theorem locally defines a differentiable function $p^M(\alpha)$:

$$\frac{\partial p^M(\alpha)}{\partial \alpha} = -\frac{\Phi_{\alpha}^M\big(p^M(\alpha), \alpha\big)}{\Phi_{p_1}^M\big(p^M(\alpha), \alpha\big)}.$$

[6]Derivations can be found in the Appendix.

A local semi-separating continuation below the cutoff requires that, for $\alpha < 1$ sufficiently close to one,

$$p^M(\alpha) < \widehat{p}^M.$$

Since $\Phi^M_{p_1} > 0$, this requires

$$\Phi^M_{\alpha} < 0$$

on the cutoff-connected semi-separating branch. The IFT can equivalently be written as:

$$\frac{\partial \alpha^M(p_1)}{\partial p_1} = -\frac{\Phi^M_{p_1}\big(p_1, \alpha^M(p_1)\big)}{\Phi^M_{\alpha}\big(p_1, \alpha^M(p_1)\big)} > 0.$$

Thus, locally along the selected cutoff-connected branch, a lower pooling price is accompanied by a decreased pooling probability and limits the scope of manipulation as trade probability increases. This local characterization does not impose uniqueness or global monotonicity of the semi-separating equilibrium correspondence.

Once the decision of the low-type firm with respect to choosing a pooling probability α is described as a function of the initial price set by the high-type, it is possible to characterize the high type's pricing strategy. The problem that the high-type firm faces is depicted in equation (1). However, as the low-type may separate by price with a positive probability, the pooling feedback channel becomes active.

Along the locally selected semi-separating branch, an interior equilibrium to the high-type firm's problem satisfies the following:

$$\frac{dU^M_H(p_1, \alpha^M(p_1))}{dp_1} = \underbrace{R_1'(p_1) + R_2^{M\prime}(p_1)}_{\text{at fixed } \alpha^M} + \underbrace{p_1 \frac{\partial \lambda_1}{\partial \bar{s}_1} \frac{\partial \bar{s}_1}{\partial \alpha^M} \alpha^{M\prime}(p_1)}_{(B1)\ t=1 \text{ via } \Delta\alpha} + \underbrace{\frac{\partial R_2}{\partial \bar{s}^M_{2,H}} \frac{\partial \bar{s}^M_{2,H}}{\partial \alpha^M} \alpha^{M\prime}(p_1)}_{(B2)\ t=2 \text{ via } \Delta\alpha} = 0.$$

For an arbitrary fixed pooling probability, define $F^M(p_1, \alpha) := \frac{\partial U^M_H(p_1,\alpha)}{\partial p_1}$, which corresponds to the review channel at some fixed pooling probability. The pooling channel comprises of the two distinct effects that affected the decision across two periods. Component $B1$ describes the impact of pooling probability on the initial perceived quality of the firm and the associated willingness-to-pay. As α goes down the expected quality conditional on observing the pooling price increases resulting in higher trade probability at a given price. Component denoted by $B2$ impacts the revenues via the expected quality in the second-period. Pooling probability has a dual effect there: (i) the probability α directly enters the updating through Bayes' rule, and (ii) initial expected quality affects the expected trade λ_1 and in consequence also the scope for manipulation $\gamma(\lambda_1)$ and the expected posterior μ_2.

Given that locally $\alpha^{M\prime}(p_1) > 0$, as well as $\lambda_{1,\bar{s}_1} > 0$, $\bar{s}_{1,\alpha} < 0$ and $\bar{s}^M_{2,H,\alpha} < 0$[7], the direction of impact of components $B1$ and $B2$ on the FOC can be determined. From the above it follows that:

$$B1(p_1) < 0, \quad B2(p_1) < 0.$$

[7]At fixed λ_1, each continuation posterior expected quality is increasing in the belief μ_1: $\frac{\partial \bar{s}^M_{2,H}}{\partial \mu_1} > 0$. Furthermore, $\mu_{1,\alpha} < 0$, and $\lambda_{1,\alpha} = \lambda_{1,\bar{s}_1}\bar{s}_{1,\alpha} < 0$. Appendix A.6 gives $\frac{\partial \bar{s}^M_{2,H}}{\partial \lambda_1} = D_M(\lambda_1, \alpha) > 0$, and the chain rule implies that at fixed price p_1: $\frac{\partial \bar{s}^M_{2,H}}{\partial \alpha} = \frac{\partial \bar{s}^M_{2,H}}{\partial \mu_1}\mu_{1,\alpha} + D_M(\lambda_1, \alpha)\lambda_{1,\alpha} < 0$.

Once the pooling channel becomes active it adds two downward pushing components to the high type's marginal payoff with respect to period one price.

A selected semi-separating equilibrium is therefore characterized by a pair (p_1^M, α^M) such that $p_1^M < p_1^{NM}$ and $\alpha \in (0,1)$, which satisfies two conditions:

1. $\Phi^M(p_1^M, \alpha^M) = 0$,
2. $F^M(p_1^M; \alpha^M) + B1(p_1^M) + B2(p_1^M) = 0$.

The first condition makes the low type indifferent between pooling and separating by playing optimal static price $p_1^{static}(s_L)$. The second selects the price preferred by the high type along the local semi-separating branch.

4.4.3 The separating price

Finally, before moving on to equilibrium characterization, lets consider the case of a complete price separation. It requires such strategy profile in which both types choose different initial prices that unambiguously signal the type and none of the types has a profitable deviation. In the context of the current model, the public message of a specific type can always be replicated with positive probability by the other one. It stems from the fact that the review transmission is neither a "good news" or "bad news" model, i.e. both types can obtain either. Any type can mimic the other's price and subsequent disclosure strategy without generating a history outside the support of the mimicked path.

Therefore, any of the types can mimic the other one by choosing an off equilibrium path strategy. The only strategy profile which can be supported as an equilibrium requires that both types are indifferent between following own strategy and choosing the other type's price. The best payoff that the low-type firm can secure when separating is the one resulting from choosing a static optimal price twice, $U_L^S = 2\kappa s_L$. This constitutes a lower bound on the low type's utility, i.e. it can always be achieved regardless of the strategy of the high-type firm. Since both types need to be indifferent, it follows that the high-type firm also obtains the same payoff, $U_H^S = U_L^S = 2\kappa s_L$, by choosing some separating price p_1^H. Given the assumption on off-path beliefs, none of the types would have a profitable deviation from such strategy.

However, such profile is strictly payoff-dominated for both types by the best pooling equilibrium $p_1^{static}(\bar{s}_1)$. Such pooling gives the initial period payoff of $\kappa\bar{s}_1$ and continuation revenue of at least κs_L. Under the maintained pessimistic off-path price beliefs, every off-path deviation can yield at most $2\kappa s_L$. The pooling profile is therefore a PBE and strictly payoff-dominates every fully separating outcome. Such payoff-dominated strategy profile is then discarded in favor of the payoff-dominating strategy.

Lemma 4 (Payoff-dominated full separation)**.** *If a fully separating PBE exists, it gives both firm types the same total payoff:*

$$U_H^S = U_L^S = 2\kappa s_L.$$

There also exists a pooling PBE in which both types charge $p^{\text{static}}(\bar{s}_1)$*, and obtain strictly higher payoff. Consequently, any fully separating PBE is strictly payoff-dominated for both firm types and is not selected.*

Proof is left for Appendix A.11.

4.4.4 Equilibrium characterization

The previous sections identified the conditions which determine the type of equilibrium faced in the game under manipulation regime. Recall that the pooling occurs when the resulting high-type price is weakly above the cutoff:

$$\widehat{p}^M := \inf\left\{p_1 : \Phi^M(p_1, 1) \geq 0\right\}.$$

Since at the cutoff pooling probability $\alpha^M = 1$ and the pooling IC gap is continuous and strictly increasing at full pooling, it is possible to express the cutoff as a function of itself and primitives:

$$\widehat{p}^M = \frac{\kappa[2s_L - \bar{s}^M_{2,L}(\widehat{p}^M)]}{1 - G(\frac{\widehat{p}^M}{\bar{s}_1})}.$$

If $\Phi^M(0,1) \geq 0$, then the cutoff is given by $\widehat{p}^M = 0$. Hence, the cutoff $\widehat{p}^M$ can be uniquely defined as an implicit function of primitives:

$$\widehat{p}^M = \widehat{p}^M(\mu_0, s_L, s_H, G)^8.$$

Secondly, the direction of the manipulation-induced price change is determined by the comparison between λ_1^{NM} and λ^*, where at $\alpha^{NM} = 1$ the latter object is constant. Using the definition of λ_1^{NM} and λ^*, as well as the invertibility of function G, the knife-edge case separating the two price responses can be rewritten as:

$$p_1^{NM} = \bar{s}_1 G^{-1}\left(\frac{\sqrt{1-\bar{s}_1}}{1+\sqrt{1-\bar{s}_1}}\right).$$

Both sides are determined by the primitives. When the LHS is greater it results in $\lambda_1^{NM} < \lambda^*$ and creates a price hike pressure, and vice-versa. If the equality holds, it results in $p_1^M = p_1^{NM}$.

Proposition 3 (Equilibrium characterization)**.** *Let denote the first-period price set by high-type as p_1^M. Under the full-pooling regularity condition, the following PBE forms can arise for admissible primitives:*

1. *A full-pooling PBE, with $p_1^M \in \left[\widehat{p}^M, p^{\text{static}}(\bar{s}_1)\right)$, and pooling probability $\alpha^M = 1$. Then, the high-type-preferred price satisfies:*
   - *(a) If $p_1^{NM} = 0$, the unique high-type preferred full-pooling price is $p_1^M = 0$.*
   - *(b) For $p_1^{NM} > 0$, the unique high-type preferred full-pooling price satisfies:*
     - *i. $p_1^{static}(\bar{s}_1(\alpha^M)) \geq p_1^M > p_1^{NM}$ if $\lambda_1^{NM} < \lambda^*$;*
     - *ii. $p_1^M = p_1^{NM}$ if $\lambda_1^{NM} = \lambda^*$;*
     - *iii. $\widehat{p}^M \leq p_1^M < p_1^{NM}$ if $\lambda_1^{NM} > \lambda^*$.*
2. *A semi-separating PBE on a local cutoff-connected branch, with $p_1^M < \widehat{p}^M$, $\alpha^M \in (0,1)$ and $\Phi^M(p_1^M, \alpha^M) = 0$. On a separating branch the low-type sets price $p_1^{static}(s_L)$. Any interior high-type-preferred pair (p_1^M, α^M) satisfies:*

$$F^M(p_1^M; \alpha^M) + B_1(p_1^M) + B_2(p_1^M) = 0.$$

[8] Closed form solution is not possible with general form of function $G(\cdot)$. Appendix proves that the cutoff exists and is unique.

If the fully separating PBE with a corresponding $\alpha^M = 0$ exists, by Lemma 4 it is strictly payoff-dominated for both types, therefore never selected. Among the pooling and semi-separating PBE, a high-type preferred equilibrium is selected.

The review disclosure is characterized by Lemma 3. The beliefs following prices and disclosure are updated according to Bayes' rule. The second period strategies follow characterization in Corollary 1.

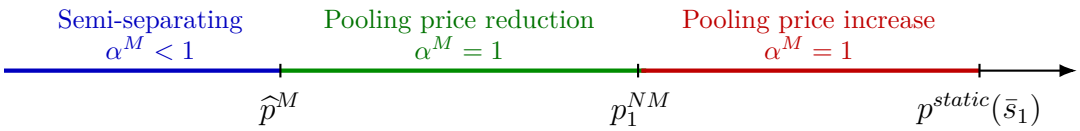
Proof is left for Appendix A.12.

Appendix A.10 shows that the parameter regions satisfying Assumption 2 and the full-pooling regularity condition, and the parameter region supporting pooling and semi-separating PBE are jointly non-empty.

In the fully pooling equilibrium, the high type's pricing problem is defined on the compact interval:

$$\left[\widehat{p}^M, p^{\text{static}}(\bar{s}_1)\right].$$

The upper bound gives the one-shot optimal pooling price. Any price above would effectively lower the revenues of the high-type. Continuity of the high type's payoff and the regularity condition therefore guarantee the existence of a unique full-pooling maximizer on this interval. Manipulation may change the equilibrium price without changing the form of the pricing equilibrium. The learning slope threshold determines the direction of the price response. The price increase occurs when learning slope is steeper without manipulation, and it necessarily preserves the pooling. The reduction compared to p_1^{NM} occurs when the learning slope is steeper under manipulation. The equilibrium structure is unchanged provided that the price remains weakly above the pooling cutoff. Once the switch to manipulation regime induces a price decrease below the cutoff $\widehat{p}^M$, the low type pools with an interior probability and the pooling-feedback channel becomes active. The axis below represents the low-type's strategies as a function of high-type's initial equilibrium price choice.



The following section studies how changes in expected quality, distance between types and demand distribution affect the local price forces. Sensitivity analysis as well as welfare analysis and comparison of manipulation modes focus on the full-pooling case. Such approach allows for isolating the impact of manipulation on informativeness and resulting prices and welfare they determine.

4.5 Sensitivity analysis

To build an understanding of the impact of manipulation on outcomes, this section focuses on the connection between primitives and the resulting local price forces. Thus far, the analysis showed that in the full-pooling case the manipulation can generate both upward and downward pressure on prices, depending on the marginal informational value of additional trade. Whether a firm decides to increase the likelihood of a review arrival depends on its impact on the posterior belief, the cost of it and the scope for manipulation. The first part of this

subsection tackles the value and cost of review generation. Then, the connection to the primitives of the model, such as firm qualities and the demand function, is built. The scope for manipulation is analysed in section 6, where the two manipulation technologies are compared.

For $k \in \{NM, M\}$, recall the FOC of the high-type's problem under the full-pooling, $F^k(p_1)$:

$$R_1'(p_1) + \kappa D_k(\lambda_1)\lambda_{1,p1}(p_1) = 0.$$

Rearranging the above yields:

$$\frac{R_1'(p_1)}{-\lambda_{1,p_1}(p_1)} = \kappa D_k(\lambda_1),$$

where the LHS summarizes the current revenue loss to marginally increase the likelihood of a review generation, whereas RHS states the continuation value of that additional probability. For the interior equilibrium to be sustained, the high-type firm increases the likelihood of generating a review until the marginal value is equal to the extra cost.

The LHS can be further rewritten using the definition of the hazard rate:

$$\frac{R_1'(p_1)}{-\lambda_{1,p_1}(p_1)} = \bar{s}_1 \left[\frac{1}{h(r)} - r\right], \quad r = \frac{p_1}{\bar{s}_1}.$$

As for the log-concave distribution $G(\cdot)$ the hazard rate is increasing in r, the cost of generating reviews decreases in p_1. Demand sensitivity determines how costly it is to use price as an instrument for producing reviews. A higher local hazard means that a small price cut increases the likelihood of a review generation relative to the current-revenue loss.

Using the above identity, the intertemporal optimality can be rewritten as:

$$\bar{s}_1 \left[\frac{1}{h(r)} - r\right] = \kappa D_k(\lambda_1).$$

The RHS is constant in the no-manipulation case, and it is increasing in λ_1 under manipulation as $D_{M,\lambda_1} > 0$. The LHS does not depend on whether manipulation is available, but depends on the hazard rate at p_1. Since the trade probability decreases in price, under manipulation both RHS and LHS decrease in price p_1. Higher price is then connected to lower marginal cost of increasing review generation probability. However, at higher price the gain from such review marginally decreases conditional on manipulation.

The intertemporal problem described above determines the scale of price change aimed at generating reviews with and without manipulation. The next step is to understand how manipulation-induced marginal price force is affected by the underlying primitives. Recall the expression for ΔF:

$$\Delta F = \kappa \lambda_{1,p_1}(p_1^{NM}) \left[D_M(\lambda_1^{NM}) - D_{NM}\right] \iff \Delta F = -\frac{\kappa g(r^{NM})}{\bar{s}_1} \left[D_M(\lambda_1^{NM}) - D_{NM}\right].$$

As established earlier, the sign of ΔF determines the direction of initial price shift conditional on the full-pooling. The expression outside the bracket is strictly negative. Therefore, the direction is determined by the sign of the bracket. Using the definition of learning slopes with and without manipulation the above can be further rewritten:

$$\Delta F = \underbrace{-\frac{\kappa g(r^{NM})}{\bar{s}_1}\left[\frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}_1}\right]}_{\text{Magnitude of price force}}\underbrace{\left[\frac{1}{(1-(1-G(r^{NM}))\bar{s}_1)^2}-\frac{1}{1-\bar{s}_1}\right]}_{\text{Determines the sign}}. \tag{2}$$

The above expression separates the source of direction of the price shift from the component scaling the magnitude of the local price force. As established in Proposition 2, the last bracket determines the sign through the comparison of demand at price p_1^{NM} to the benchmark λ^*. Depending on the benchmark objects, the magnitude is scaled by the local density of demand and the distance between qualities. As $|\Delta F|$ measures the deviation of the marginal payoff from zero at the no-manipulation benchmark price p_1^{NM} under manipulation, a higher $|\Delta F|$, *ceteris paribus*, creates a larger manipulation-induced local price force at the no-manipulation benchmark price.

Proposition 4 (Manipulation-induced price force)**.** *Suppose that $p_1^{NM} > 0$ and consider the full-pooling equilibrium. The magnitude and direction of the manipulation-induced marginal price force follow from equation* (2)*. For $\lambda_1^{NM} \neq \lambda^*$, holding all other terms in equation (2) fixed:*

1. *Magnitude of the local price force:*
   (a) *higher local density $g(r^{NM})$ scales up the price force $|\Delta F|$;*
   (b) *larger distance between firm types Δs increases $|\Delta F|$, scaled by $(\Delta s)^3$. As $\Delta s \to 0$, the manipulation-induced informational price force vanishes.*
2. *Direction of the local price force:*
   (a) *higher benchmark demand $\lambda_1^{NM} = 1 - G(r^{NM})$ increases local downward price force.*
   (b) *higher expected quality $\bar{s}_1$ raises the threshold $\frac{\partial \lambda^*}{\partial \bar{s}_1} > 0$, limiting the scope for a price decrease.*

Proof is left for Appendix A.13.

At higher local density $g(r^{NM})$ a marginal decrease in price adds greater likelihood of generating a review, while the revenue loss is smaller due to higher relative increase in demand. Conditional on an unchanged expected quality $\bar{s}_1$, greater distance between types' qualities Δs increases the scale of a local price force. The informational gain for the high-type is greater when the low-type pulls the expected quality in $t = 2$ down. Additionally, greater wedge between the types strengthens the informational load of a review.

Consumer valuation distribution $G(\cdot)$ affects the demand at a given price and expected quality. At fixed price and expected quality, higher level of demand at a benchmark price increases the downward pressure on the price response, as manipulation becomes less effective and the scope for censoring falls in the expected trade probability. Finally, as the expected quality goes up the threshold λ^* increases. Higher expected quality makes the downward price shift more costly for the high-type. A larger benchmark trade probability λ_1^{NM} is required to make the marginal increase in trade probability sufficiently informative to offset the loss of initial revenues.

5 Welfare

The previous section characterized the equilibrium of the game and discussed the effect of primitives on the outcome. With the focus on the pooling PBE, this section discusses the effect of resulting prices on the surplus of consumers and total efficiency of the market. A natural conjecture is to think of review manipulation as an undesired feature in the relation between a firm and its customers. However, as section 4 showed, under some circumstances the existence of manipulation may put a downward pressure on price compared to the no-manipulation benchmark.

Define $r_t := \frac{p_t}{\bar{s}_t}$. Consider a consumer and total surplus, respectively, in period t defined as:

$$CS_t(p_t, \bar{s}_t) := \int_{r_t}^{1} (v\bar{s}_t - p_t)g(v)dv = \bar{s}_t \int_{r_t}^{1} (v - r_t)g(v)dv, \quad \text{and} \quad TS_t(p_t, \bar{s}_t) = \bar{s}_t \int_{r_t}^{1} vg(v)dv,$$

where the total surplus is a sum of consumer surplus and firm's revenues. Then, each of the ex-ante surpluses over the entire duration of the game is a sum of two periods. Define the consumer surplus and total surplus, respectively, for a regime $k \in \{M, NM\}$:

$$CS^k := \mathbb{E}^k \left[\sum_{t=1}^{2} CS_t(p_t^k, \bar{s}_t^k) \right], \quad \text{and} \quad TS^k := \mathbb{E}^k \left[\sum_{t=1}^{2} TS_t(p_t^k, \bar{s}_t^k) \right].$$

In expectation, consumer surplus is greater than zero only for consumers with valuation v such that $v > \frac{p}{\bar{s}}$. When the price decreases, at a fixed expected quality, the expected consumer surplus goes strictly up. More types of consumer can participate in trade, while those that were already buying at a higher price can enjoy a bigger spread between their valuation v and the final price. The total surplus is a sum over consumer's surplus and firm's revenues. When price changes the surplus becomes a transfer between consumers and firm. However, the shift in trade probability can either expand when price drops or shrink the market when price hikes. In the second period, the optimal monopoly price is proportional to posterior expected quality. Since both consumer and total surplus are linear in expected quality, their ex-ante period two values are identical across the regimes by Bayes plausibility.

Proposition 5 (Manipulation welfare effect)**.** *Suppose $p_1^{NM} > 0$, the regularity condition holds, and manipulation preserves the full-pooling equilibrium structure. Using Proposition 2, when manipulation results in the initial price decrease, $p_1^M < p_1^{NM}$, the ex-ante consumer and total surpluses increase compared to the no-manipulation case:*

$$\lambda_1^{NM} > \lambda^* \implies p_1^M < p_1^{NM} \implies CS^M > CS^{NM}, \quad \textit{and} \quad TS^M > TS^{NM}.$$

Conversely, if manipulation induces a price hike, $p_1^M > p_1^{NM}$, the ex-ante consumer and total surpluses decrease compared to the no-manipulation case:

$$\lambda_1^{NM} < \lambda^* \implies p_1^M > p_1^{NM} \implies CS^M < CS^{NM}, \quad \textit{and} \quad TS^M < TS^{NM}.$$

If manipulation does not affect the price then:

$$\lambda_1^{NM} = \lambda^* \implies p_1^M = p_1^{NM} \implies CS^M = CS^{NM}, \quad \textit{and} \quad TS^M = TS^{NM}.$$

Proof. At a fixed expected quality, both measures of surplus are decreasing in every interior price, calculated using Leibniz rule:

$$\frac{\partial CS_t(p_t, \bar{s}_t)}{\partial p_t} = -\left[1 - G\left(\frac{p_t}{\bar{s}_t}\right)\right] < 0, \quad \text{and} \quad \frac{\partial TS_t(p_t, \bar{s}_t)}{\partial p_t} = -\frac{p_t}{\bar{s}_t} g\left(\frac{p_t}{\bar{s}_t}\right) < 0.$$

Therefore, the initial price decrease at fixed expected quality raises the first period ex-ante consumer surplus as some obtain a strictly greater surplus than under the no-manipulation price, while the trade probability expands and more consumers can have positive surplus. The opposite happens when manipulation availability induces the price increase.

The change in price results in a transfer from or to a firm. When the price decreases the probability of trade goes up adding more customers with ex-ante positive surplus. Hence, in response to a price cut in period one the initial total surplus in the market increases compared to the benchmark under no-manipulation. The opposite effect occurs when the initial price increases. The additional revenue for the firm comes from a surplus transfer from customers. And since the expected quality is fixed under full-pooling, some consumers are driven out of the market and the total surplus goes down.

Regardless of the direction of price change, the second period surpluses remain neutral. As per Corollary 1, the optimal price in period two is given by:

$$p_2^k = r^* \bar{s}_2^k.$$

The firm always sets the optimal static price in period two at a realized expected quality. Then the purchase cutoff is equal to r^* after every public history. Therefore, the trade probability remains unchanged, and only the amount of ex-ante surplus differs in the expected quality of the firm. From the definition of both CS_t^k and TS_t^k it can be noted, that they are both linear in the expected quality. By Bayes plausibility, the expectation of the realization is equal to ex-ante expectation:

$$\mathbb{E}^k[\bar{s}_2^k] = \mathbb{E}^k\left[\mathbb{E}[s \mid p_1^k, m]\right] = \mathbb{E}[s] = \bar{s}_1.$$

This implies that when the price set in period two differs only by the linear scaling factor $\bar{s}_2^k$, the surplus measures are ex-ante equal:

$$\mathbb{E}^M[CS_2^M(p_2^M, \bar{s}_2^M)] = E^{NM}[CS_2^{NM}(p_2^{NM}, \bar{s}_2^{NM})],$$

and

$$E^M[TS_2^M(p_2^M, \bar{s}_2^M)] = E^{NM}[TS_2^{NM}(p_2^{NM}, \bar{s}_2^{NM})].$$

Conditional on full-pooling equilibrium, manipulation and associated initial price change affect the ex-ante welfare outcomes only in the initial period. □

6 Fake reviews

Censorship of unfavorable reviews is not the only method to manipulate the beliefs of consumers. In many setting removing unfavorable information may be more difficult for the firm than generating fake reviews that praise its product. Changing the manipulation technology from censoring to creating fakes alters consumers' inference regarding the underlying quality. After all, up to this point the low-type firm was able to hide the bad signal and blur the

inference. But a good review was strictly increasing the equilibrium belief as its likelihood was higher for the high-type. However, once adding fake reviews becomes a possibility, the pool of good reviews is contaminated by fakes, and the good signal loses its credibility. This section studies how the information problem changes with the mode of manipulation and what are its equilibrium implications.

Consider an extension to the model, under which firm cannot pick a probability γ of hiding a review generated by consumer in period one, i.e. $\gamma = 0$. Instead, it can send any public message $m \in \{0, 1, \varnothing\}$ conditional on no trade occurring in period one. It can be noted that if the firm generated a fake review that was published along with a real review written by a consumer, such manipulation would be obvious to the second period buyer. Therefore, it is assumed that fake review can be generated only when $y = \varnothing$.

When a trade occurs and review is non-empty, $y \in \{0, 1\}$, the review is transmitted truthfully, $m = y$. Conditional on no trade in the initial period there is no review, $y = \varnothing$. Denote by $\rho_{i,1}$ the probability that type i fabricates a positive review and let $\rho_{i,0}$ denote the probability that it generates a negative review. With probability $1 - \rho_{i,1} - \rho_{i,0}$, the firm truthfully sends message $m = \varnothing$.

Throughout this section, the truth-leaning Assumption 1 is applied to the fake reviews generation. Let $\rho^0 \equiv 0$ denote a truthful strategy profile. For $j \in \{0, 1\}$, assume that the following condition holds:

$$\mu_2(p_1, j, \rho^*) \leq \mu_2(p_1, \varnothing, \rho^0) \quad \Longrightarrow \quad \rho^*_{i,j} = 0,$$

which is fake review generation equivalent of Assumption 1. The firm only fabricates reviews when it strictly increases the associated belief compared to the truthful strategy.

Lemma 5 (Fake review strategy)**.** *In any truth-leaning pooling PBE, fake review strategy is such that public message m satisfies $\mu_2(m = \varnothing) = \mu_2(m = 1)$. Fabricating bad reviews is payoff-dominated. Moreover, only the low-type produces fake good reviews with positive probability, $\rho_L \geq 0$, $\rho^*_H = 0$, where:*

$$\rho^*_L(\lambda_1) = \frac{\lambda_1(s_H - s_L)}{1 - \lambda_1 + \lambda_1 s_H}.$$

Proof is left for Appendix A.14.

It is always payoff-dominated to generate a bad review. Since the likelihood of an honest bad review is greater for the low-type it strictly lowers the posterior belief. When positive fake reviews are generated, they shift the posterior beliefs on-path. For the fake generation strategy to constitute a part of a PBE, the resulting posteriors after empty and good message must be equal. Otherwise, there would exist a profitable deviation to either generate less or more fakes. Similarly to the case of concealment, the high-type does not benefit from generating fake positive reviews. The low-type may increase its own fake review generation probability to preserve $\mu_2(m = \varnothing) = \mu_2(m = 1)$. Such strategies only reallocate the probability between the two message yielding the same belief, and continuation payoffs. Under the truth-leaning assumption, once the high-type does not benefit from untruthful messaging strategy, it chooses $\rho_H = 0$.

Corollary 4. *Probability ρ^*_L of generating fake good review increases in λ_1, but the probability of no honest review declines in the trade probability.*

When the probability of trade goes up consumers expect to see an empty message less often. This allows the low-type to fabricate reviews with higher probability, but the likelihood of such opportunity decreases in λ_1.

Once the optimal manipulation strategy under the fake reviews regime is established, consider its impact on the equilibrium pricing. Let $\bar{s}_2^k(\lambda_1)$ denote the expected posterior quality for each regime in $k \in \{F, C\}$, corresponding to fake reviews and censoring, respectively. Then, let define the associated learning slopes:

$$D_k(\lambda_1) = \frac{\partial \bar{s}_{2,H}^k(\lambda_1)}{\partial \lambda_1}.$$

As the focus still lies on the full-pooling case, the direction of price adjustment under manipulation regime k depends on the comparison between the associated manipulation and the no-manipulation learning slopes.

As the manipulation enters the optimization problem via the expected quality $\bar{s}_2^k$, the impact of the two manipulation technologies in a full-pooling PBE differs only through the learning slope $D_k(\lambda_1)$. Evaluating learning slopes under both regimes at the no-manipulation benchmark price and trade probability results in:

$$D_C(\lambda_1^{NM}) = \frac{\mu_1(1-\mu_1)^2(s_H - s_L)^3}{\bar{s}_1(1-\lambda_1^{NM}\bar{s}_1)^2}, \quad D_F(\lambda_1^{NM}) = \frac{\mu_1(1-\mu_1)^2(s_H - s_L)^3}{(1-\bar{s}_1)(1-\lambda_1^{NM}(1-\bar{s}_1))^2}.$$

To evaluate whether under fake reviews regime the manipulation creates a downward pushing price force compare:

$$D_F(\lambda_1^{NM}) > D_{NM} \iff \lambda_1^{NM} > \lambda_F^* := \frac{1}{1+\sqrt{\bar{s}_1}},$$

and additionally denote the threshold λ^* showed in section 4 as λ_C^*.

Proposition 6 (Manipulation technology and price response)**.** *Suppose that $p_1^{NM} > 0$, $\bar{s}_1 > \frac{1}{2}$, the full-pooling regularity condition holds, and both manipulation technologies preserve full pooling. For $\bar{s}_1 > \frac{1}{2}$, the manipulation regime specific thresholds satisfy $\lambda_F^* < \lambda_C^*$. Using the argument of Proposition 2, the direction of price change depends on the relation of λ_k^* to λ_1^{NM}:*

1. *If $\lambda_F^* < \lambda_C^* < \lambda_1^{NM}$, price decreases under both regimes.*
2. *If $\lambda_F^* < \lambda_1^{NM} < \lambda_C^*$, price increases under censoring, and decreases under fake reviews.*
3. *If $\lambda_1^{NM} < \lambda_F^* < \lambda_C^*$, price increases under both regimes.*

For $\lambda_k^* = \lambda_1^{NM}$ price under manipulation remains unchanged from benchmark.

The welfare implications of initial price change are as described in Proposition 5.

Proof is left for Appendix A.15.

The two technologies affect the inference of consumers differently. Under censorship the bad and empty message are pooled together, whereas a good review is always a positive signal. On the opposite end, the fake reviews regime diminishes the impact of a good review, while

the bad review becomes more informative. When $\bar{s}_1$ is greater than half, good reviews are relatively more likely to be generated compared to the bad ones. Because of this, censoring is less effective as consumers expect to observe a good review with greater probability. This raises the threshold λ_C^* compared to λ_F^* and reduces the incentive to decrease the price under censorship. Conversely, when the low-type firm can fabricate good reviews, it forces the high-type to react sooner to limit the scope for manipulation and generate greater separation from the low-type via reviews. Therefore, the direction of price response under both regimes may differ depending on relative thresholds and the expected quality.

Remark 1. *If $\bar{s}_1 < \frac{1}{2}$ and conditional on sufficiency of the regularity condition under fake reviews, the ordering of the technologies is reversed, and the price-force characterization applies in the opposite direction. At $\bar{s}_1 = \frac{1}{2}$, the two technologies have the same price effect in equilibrium.*

Remark 2. *The learning slopes $D_k(\lambda_1^{NM})$can determine the relative marginal pricing effect of the two manipulation regimes. Since:*

$$\Delta F^F - \Delta F^C = \kappa \lambda_{1,p_1}(p_1^{NM}) \left[D_F(\lambda_1^{NM}) - D_C(\lambda_1^{NM}) \right],$$

and $\lambda_{1,p_1} < 0$, the relation of D_F to D_C determines the relative shift in the high type's marginal pricing response. Conditional on the direction of the price change, the learning slopes identify which technology has stronger marginal effect.

7 Conclusions

The analysis presented throughout the article shows how context specific the impact of manipulation is. All else fixed, manipulation lowers the informativeness of a review and renders feedback left by others less meaningful. However, it does not exist in vacuum, and availability of the manipulation technology affects the dynamics of the firm problem. In a pooling equilibrium, the possibility of one type manipulating the reviews may lead to a price decrease and in the process increase the consumer surplus and total welfare. However, if the learning slope under manipulation is flatter than in the no-manipulation benchmark the manipulation can result in higher prices, falling trade probability and a decrease in market efficiency. When the prices set by both types are not perfectly aligned, the problem becomes more complex. Additional bits of information are passed on through the price alone, while the effectiveness of manipulation also varies in the likelihood of pooling. For semi-separating equilibrium to arise price needs to decrease, but its welfare implications remain an open question.

The results also indicate that the technology of manipulation matters for pricing strategies and associated welfare outcomes. Censorship and fabricating reviews affect the beliefs differently. When a firm can remove existing evidence, Lemma 3 indicated that the low-type firm manipulates by censoring bad reviews and in the process decreases the posterior belief conditional on an empty message. When a firm can create fakes, its manipulation strategy depends on the lack of evidence, which can be then fabricated. Lemma 5 shows that the low-type firm generates positive fake reviews, and by doing so, it lowers the credibility of a good signal. Depending on the learning slopes, the benchmark trade and expected quality, a high-type firm may react by either lowering a price to increase the separation via reviews; or it may move the price towards the static optimum to earn higher payoff in period one. Under censorship, increasing trade probability limits the scope for concealment. On the other hand,

the probability that a low-type firm produces a fake review conditional on no review increases with the trade probability. But as the likelihood of trade increases it is less likely that it has an opportunity to fabricate. Under full pooling, the welfare impact of technology depends whether it eventually leads to a price decrease. Depending on the initial expected quality, the two technologies lead to different thresholds for a price cut. Their welfare effects depend on underlying market conditions, and technologies may affect the sign of the welfare outcomes differently. The analysis provides an argument to consider the impact of different modes of manipulation when regulating environments such as online platforms.

The complexity of the price-information environment leaves some potential avenues for future research. One of the natural extensions of this framework is to study the infinite horizon problem. How does the price dynamics evolve then – subdued prices early on to produce information before hiking or rather stable over-time. Additionally, in such environment learning pace would shift with changing beliefs, creating another force in the equilibrium.

Another possible extension is to relax the assumption that every buyer leaves an honest review. Depending on the factors, whether prices or expectations, affecting the review, this would reshape the disclosure problem and create space for new trade-offs in the model. Finally, the current model focuses on two manipulation technologies, but they are not available simultaneously. Allowing for some degree of overlap may lead to a different information disclosure/fabrication strategies, and affect the equilibrium pricing incentives.

The review manipulation on a market cannot be evaluated solely based on the informational distortions it generates. When information is endogenous, the informational effects mix with prices and market forces, and create new trade-offs. A firm may react by adjusting prices and the informational distortions have a ripple effect on the market outcomes. Regulation of review environments and online platforms should account for the credibility of available information and the market response induced by the manipulation technology.

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A Proofs

A.1 Proof of Lemma 1

Proof. There are three candidate strategies to consider: 1) pooling 2) separating, and 3) semi-separating. In a one-shot game the only source of information is the price set. Moreover, since the game finishes immediately after period one, the objective of both types is to maximize period one revenues.

Consider the best pooling price that both types may set. It solves the FOC with respect to the maximization problem at a shared belief μ_0, which results in the expected quality $\bar{s} = \mu_0 s_H + (1-\mu_0)s_L$. Such price is given by:

$$p^{static}(\bar{s}) = r^* \bar{s},$$

and results in expected revenues equal to

$$R^{static}(\bar{s}) = \kappa \bar{s}.$$

Due to assumed off-path beliefs, any deviation from that pricing strategy results in revenues equal to at most κs_L, hence none of the types has a profitable deviation as $s_L < \bar{s}$. Therefore such pooling constitutes a PBE. Moreover, pooling at any other price is payoff-dominated.

Then, consider any separating equilibrium. As both types necessarily choose different prices, let V_H denote the revenue following high-type's price, and V_L the revenue associated with the price set by the low type. For such pair of prices to be a PBE, there cannot exist any profitable deviation. Any type can achieve the revenue of the opponent by choosing her price. Therefore, the incentive compatibility requires that:

1. $V_H \geq V_L$,
2. $V_L \geq V_H$.

This implies $V_H = V_L$, i.e. any separating candidate price must rely on exact indifference. The low-type's separating price can generate revenues equal to at most κs_L. Since κs_L is also the best possible payoff off-path due to the assumed beliefs, both types obtain such payoff in a separating PBE for it to be sustained with no profitable deviations. However, it is payoff-dominated by the pooling PBE.

Finally, consider semi-separating strategies. Any equilibrium in which the low-type separates with positive probability results in the same considerations as above. For it to be a PBE, the separating and pooling branch must yield the same expected payoff to the low-type of at most κs_L. Therefore, any such PBE is payoff-dominated. If the high-type separates with a positive probability, the payoff on the pooling branch is at most equal to $\kappa \tilde{s}$, where $\tilde{s} < \bar{s}$ as the probability weight of the type s_H is less than μ_0. For it to be a PBE, the separating branch must make both types indifferent and yield $\kappa \tilde{s}$ as well. Therefore, both types obtain payoff of at most $\kappa \tilde{s} < \kappa \bar{s}$, and such PBE is payoff-dominated. □

A.2 Continuation beliefs under no manipulation

Denote:

$$\bar{s} := \bar{s}_1^{NM} = \mu_0 s_H + (1-\mu_0)s_L, \qquad \Delta s := s_H - s_L,$$

and let

$$\lambda_1 = \lambda(p_1, \bar{s}).$$

Under full pooling equilibrium, the first-period price does not affect the belief in $t = 1$. In the absence of manipulation, posterior beliefs following a positive, a negative and no review are, respectively:

$$\mu_2(1) = \frac{\mu_0 s_H}{\bar{s}}, \qquad \mu_2(0) = \frac{\mu_0(1 - s_H)}{1 - \bar{s}}, \qquad \mu_2(\varnothing) = \mu_0.$$

The no review posterior preserves the prior as there is no new information to use.

The consumers assign corresponding second-period expected qualities:

$$\bar{s}_2(1) = \bar{s} + \frac{\mu_0(1 - \mu_0)(\Delta s)^2}{\bar{s}}, \quad \bar{s}_2(0) = \bar{s} - \frac{\mu_0(1 - \mu_0)(\Delta s)^2}{1 - \bar{s}}, \quad \bar{s}_2(\varnothing) = \bar{s}.$$

Conditional on the firm being the high type, its expected second-period reputation is:

$$\bar{s}^{NM}_{2,H}(\lambda_1) = (1 - \lambda_1)\bar{s} + \lambda_1 s_H \bar{s}_2(1) + \lambda_1(1 - s_H)\bar{s}_2(0) = \bar{s} + \lambda_1 \frac{\mu_0(1 - \mu_0)^2(\Delta s)^3}{\bar{s}(1 - \bar{s})}.$$

Then, define:

$$D_{NM} := \frac{\mu_0(1 - \mu_0)^2(\Delta s)^3}{\bar{s}(1 - \bar{s})},$$

to be a no-manipulation learning slope, i.e. denotes how the expected quality conditional on being the high-type changes in additional trade probability λ_1. Then, the expression $\bar{s}^{NM}_{2,H}(\lambda_1)$ can be rewritten as:

$$\bar{s}^{NM}_{2,H}(\lambda_1) = \bar{s} + D_{NM}\lambda_1,$$

where $D_{NM} > 0$.

Similarly, conditional on the firm being the low type, its expected second-period perceived quality is:

$$\bar{s}^{NM}_{2,L}(\lambda_1) = (1 - \lambda_1)\bar{s} + \lambda_1 s_L \bar{s}_2(1) + \lambda_1(1 - s_L)\bar{s}_2(0) = \bar{s} - \lambda_1 \frac{\mu_0^2(1 - \mu_0)(\Delta s)^3}{\bar{s}(1 - \bar{s})}.$$

Then:

$$\bar{s}^{NM}_{2,L}(\lambda_1) = \bar{s} - A_L\lambda_1,$$

where

$$A_L := \frac{\mu_0^2(1 - \mu_0)(\Delta s)^3}{\bar{s}(1 - \bar{s})} > 0.$$

A.3 Proof of Lemma 2

Proof. Under full pooling and no-manipulation benchmark, the low type's expected payoff from setting price: p_1 is

$$U_L^{\text{pool}}(p_1) = p_1\lambda_1(p_1) + \kappa\bar{s}^{NM}_{2,L}(\lambda_1(p_1)).$$

Using the result of Appendix A.2, the above can be rewritten as:

$$U_L^{\text{pool}}(p_1) = p_1\lambda_1(p_1) + \kappa\left[\bar{s} - A_L\lambda_1(p_1)\right].$$

The best separating payoff is independent from price p_1 and given by:

$$U_L^{\text{sep}} = 2\kappa s_L.$$

Therefore:

$$\Phi'(p_1) = \frac{dU_L^{\text{pool}}(p_1)}{dp_1} = \lambda_1(p_1) + p_1\lambda_{1,p_1}(p_1) - \kappa A_L \lambda_{1,p_1}(p_1),$$

where $A_L > 0$ as showed in Appendix A.2, and $\lambda_{1,p_1}(p_1) < 0$ as the demand decreases in price. Moreover, on the interval $p_1 \in \left[0, p^{\text{static}}(\bar{s})\right]$, first-period revenue is weakly increasing. Then:

$$\lambda_1(p_1) + p_1\lambda_{1,p_1}(p_1) \geq 0, \quad \text{and} \quad -\kappa A_L \lambda_{1,p_1}(p_1) > 0.$$

It follows that on the relevant price interval the following holds:

$$\Phi'(p_1) > 0.$$

Continuity of Φ follows from continuity of demand and posterior expected qualities. It remains to show that the set

$$\{p_1 : \Phi(p_1) \geq 0\}$$

is non-empty. At the static monopoly price the revenues are given by:

$$R_1\big(p^{\text{static}}(\bar{s}), \bar{s}\big) = \kappa\bar{s}.$$

Furthermore, every posterior expected quality belongs to $[s_L, s_H]$, so

$$\bar{s}_{2,L}^{NM}(\lambda_1) \geq s_L.$$

Consequently,

$$\begin{aligned}\Phi\big(p^{\text{static}}(\bar{s})\big) &= U_L^{\text{pool}}\big(p^{\text{static}}(\bar{s})\big) - 2\kappa s_L \\ &\geq \kappa\bar{s} + \kappa s_L - 2\kappa s_L \\ &= \kappa(\bar{s} - s_L) > 0.\end{aligned}$$

If $\Phi(0) \geq 0$, then the cutoff is given by:

$$\widehat{p} = 0.$$

If $\Phi(0) < 0$, continuity and strict monotonicity imply that there exists a unique:

$$\widehat{p} \in \left(0, p^{\text{static}}(\bar{s})\right),$$

for which the following holds:

$$\Phi(\widehat{p}) = 0.$$

Thus, in either case,

$$\Phi(p_1) \geq 0 \quad \Longleftrightarrow \quad p_1 \geq \widehat{p}$$

on the relevant price interval. □

A.4 Proof of Proposition 1

Proof. Using the result A.2 in the Appendix, the high type's expected payoff can be written as:

$$U_H^{NM}(p_1) = p_1\lambda_1(p_1) + \kappa\bar{s}_{2,H}^{NM}(\lambda_1(p_1)) = p_1\lambda_1(p_1) + \kappa\left[\bar{s} + D_{NM}\lambda_1(p_1)\right],$$

which is equivalent to:

$$U_H^{NM}(p_1) = \kappa\bar{s} + (p_1 + \kappa D_{NM})\left[1 - G\left(\frac{p_1}{\bar{s}}\right)\right].$$

The objective function of the high-type is continuous in price p_1. Any price $p_1 \geq \bar{s}$ generates no trade and resulting in payoff $\kappa\bar{s}$. At $p_1 = 0$:

$$U_H^{NM}(0) = \kappa(\bar{s} + D_{NM}) > \kappa\bar{s}.$$

The maximization problem can be restricted to the compact interval $[0, \bar{s}]$, and a solution exists.

Suppose now that the initial price satisfies $p_1 > p^{\text{static}}(\bar{s})$. Reducing it towards $p^{\text{static}}(\bar{s})$ strictly increases first-period revenue and increases the probability of trade λ_1. Since

$$D_{NM} > 0,$$

the resulting increase in review arrival strictly improves the high type's expected second-period reputation. Hence, every price satisfying

$$p_1 > p^{\text{static}}(\bar{s})$$

is strictly dominated by $p_1 = p^{static}(\bar{s}_1)$. However, the static monopoly price itself is not optimal. At that price, the marginal effect on current revenue is zero, $R_1'\left(p^{static}(\bar{s})\right) = 0$. But the derivative with respect to the continuation revenue is lower than zero:

$$R_2'\left(p^{static}(\bar{s})\right) = \kappa D_{NM}\lambda_{1,p_1}\left(p^{\text{static}}(\bar{s})\right) < 0.$$

Therefore, the static optimal price does not maximize the revenue over the two periods:

$$\left.\frac{dU_H^{NM}}{dp_1}\right|_{p_1 = p^{\text{static}}(\bar{s})} < 0.$$

To further increase the high-type's payoff, the price can be marginally reduced:

$$0 \leq p_1^{NM} < p^{\text{static}}(\bar{s}).$$

Such price raises $R_1'(p_1^{NM})$ above zero and lowers $R_2'(p_1^{NM})$.

It remains to establish uniqueness. Define:

$$r := \frac{p_1}{\bar{s}}, \qquad c := \frac{\kappa D_{NM}}{\bar{s}} > 0,$$

and let

$$h(r) := \frac{g(r)}{1 - G(r)}$$

denote the hazard rate. As $\kappa\bar{s}_1$ is constant at full-pooling, the high type objective can be rewritten as:

$$f(r) = (r + c)[1 - G(r)].$$

The derivative of the objective then becomes:

$$f'(r) = [1 - G(r)] \left(1 - (r + c)\frac{g(r)}{1 - G(r)}\right),$$

and substituting in the hazard rate gives:

$$f'(r) = [1 - G(r)] \left(1 - (r + c)h(r)\right).$$

The derivative $f'(r)$ has the same sign as the inside of the bracket:

$$1 - (r + c)h(r).$$

Under the assumption of log-concavity of PDF g, the hazard rate $h(r)$ is non-decreasing. Since $r + c$ is strictly increasing and $h(r) > 0$, then $(r + c)h(r)$ is strictly increasing. Therefore, the first-order condition

$$(r + c)h(r) = 1$$

has at most one interior solution.

Then, the last step is to determine whether the solution exists. First, if

$$ch(0) \geq 1 \implies f'(r) \leq 0 \, \forall r^+,$$

the resulting objective decreases for every $r > 0$. Then the unique maximizer is given by $r = 0$.

Alternatively, if

$$ch(0) < 1 \implies f'(0) > 0,$$

and the solution cannot be $r = 0$. By construction the static optimum r^* satisfies:

$$r^* h(r^*) = 1,$$

and as $c > 0$ it implies:

$$(r^* + c)h(r^*) > 1.$$

Continuity and strict monotonicity imply that there exists a unique interior solution

$$r^{NM} \in (0, r^*)$$

that satisfies: $(r + c)h(r) = 1$.

Then, for either of the two cases there exists a unique solution to the high type's no-manipulation pricing problem, which satisfies:

$$0 \leq p_1^{NM} < p^{static}(\bar{s}).$$

This argument identifies the unique solution to the high type's pricing problem conditional on full pooling. Under Assumption 2, the only admissible primitives are those for which price p_1^{NM} satisfies the low type's pooling incentive constraint:

$$\Phi\left(p_1^{NM}\right) \geq 0.$$

It guarantees that at the given price the low-type has no incentive to deviate. Furthermore, as the price p_1^{NM} is chosen to maximize high-type's objective function, and at assumed off-path beliefs the high-type can obtain the payoff of at most $2\kappa s_L$ which corresponds to the best separating payoff for the low-type, it does not have any profitable deviation either. If the above inequality does not hold at $p_1 NM$ for some primitives, then such case is excluded from the analysis.

□

A.5 Proof of Lemma 3

Proof. Let consider some pooling first-period price p_1. Let $\mu_1 \in (0,1)$ denote the posterior following the pooling price, and let $\lambda_1 \in (0,1)$ denote the resulting probability of the first period trade. Finally, let denote the truthful-disclosure profile as $\gamma^0 \equiv 0$.

It can be first showed that concealing a good review cannot be an equilibrium strategy for any type. Compare the posterior following a truthful revealing of a good review and no review:

$$\mu_2(1;\gamma^0) = \frac{\mu_1 s_H}{\mu_1 s_H + (1-\mu_1)s_L} > \mu_1 = \mu_2(\varnothing;\gamma^0).$$

Then, consider a consequence of either type hiding the good review. If the low-type conceals a good review with some positive probability γ, the posteriors would adjust:

$$\mu_2(1,\gamma) = \frac{\mu_1 s_H}{\mu_1 s_H + (1-\mu_1)s_L(1-\gamma)} > \mu_2(1,\gamma^0) > \mu_2(\varnothing,\gamma^0) > \mu_2(\varnothing,\gamma) = \frac{\mu_1(1-\lambda)}{(1-\lambda)+(1-\mu_1)\lambda_1\gamma s_L}.$$

Revealing a good review is strictly better for the low-type.

Now, consider the high-type hiding reviews with some positive probability γ. For such γ to be an equilibrium strategy it would need to result in $\mu_2(1) = \mu_2(\varnothing)$, otherwise the high-type would deviate and either no censor or censor every good review. At such disclosure strategy of the high-type – absent any reaction from the low-type – $\mu_2(1) = \mu_2(\varnothing) > \mu_2(0)$. But this cannot be an equilibrium since the low-type have a profitable deviation from honest disclosure to hiding bad reviews with some probability. Then low-type sets such γ_L probability of hiding bad reviews to obtain $\mu_2(\varnothing) = \mu_2(0)$. Eventually, to sustain $\gamma > 0$ as an equilibrium strategy would result in $\mu_2(1) = \mu_2(\varnothing) = \mu_2(0)$, rendering all reviews meaningless. As $s_H > s_L$, the likelihood of high-type obtaining a good review is higher than for the low-type. The babbling equilibrium yields the expected quality $\bar{s}_1$ in period two, whereas informative disclosure of good reviews gives the high type a strictly higher expected continuation reputation. Hence, revealing all good reviews payoff-dominates hiding good reviews for the high-type.

Thus, concealing a good review reduces the firm's posterior reputation. By assumption 1, concealment happens with positive probability only when it increases the expected posterior belief:

$$\gamma^*(1 \mid s_i, p_1) = 0 \qquad \text{for } i \in \{L, H\}.$$

Now consider concealment of a bad review. Let denote type i's probability of concealing a bad review as:

$$\gamma_i := \gamma(0 \mid s_i, p_1).$$

Given (γ_H, γ_L) and conditional on the type, the likelihood of a disclosed bad review is

$$\Pr(m = 0 \mid s_i) = \lambda_1(1 - s_i)(1 - \gamma_i),$$

while the likelihood of an empty message is

$$\Pr(m = \varnothing \mid s_i) = 1 - \lambda_1 + \lambda_1(1 - s_i)\gamma_i.$$

At any pooling equilibrium price under manipulation, disclosure and concealment of a bad review must induce the same continuation payoff, otherwise any type would have a profitable deviation from such disclosure policy. Since continuation revenue is strictly increasing in the posterior expected quality, this requires the following to hold:

$$\mu_2(0) = \mu_2(\varnothing).$$

In case of $\mu_2(\varnothing) > \mu_2(0)$, a firm would benefit from concealing bad review more often. Similarly, for $\mu_2(\varnothing) < \mu_2(0)$, a firm would benefit from disclosing the bad review more often than prescribed by the equilibrium strategy.

Using the Bayes' rule formulas for posteriors after bad and empty messages gives:

$$\mu_2(0) = \frac{\mu_1(1 - s_H)(1 - \gamma_H)}{\mu_1(1 - s_H)(1 - \gamma_H) + (1 - \mu_1)(1 - s_L)(1 - \gamma_L)}, \text{ and}$$

$$\mu_2(\varnothing) = \frac{\mu_1\left[1 - \lambda_1 + \lambda_1(1 - s_H)\gamma_H\right]}{\mu_1\left[1 - \lambda_1 + \lambda_1(1 - s_H)\gamma_H\right] + (1 - \mu_1)\left[1 - \lambda_1 + \lambda_1(1 - s_L)\gamma_L\right]}.$$

Equating them and simplifying results in:

$$\frac{1 - \gamma_L}{1 - \gamma_H} = \frac{(1 - s_H)(1 - \lambda_1 s_L)}{(1 - s_L)(1 - \lambda_1 s_H)},$$

where the RHS is constant. This implies that any change in γ_H must be offset by a shift in γ_L. Using the likelihood ratio of a bad review conditional on the type gives:

$$\frac{Pr(0|s_H)}{Pr(0|s_L)} = \frac{(1 - s_H)(1 - \gamma_H)}{(1 - s_L)(1 - \gamma_L)},$$

thus the likelihood is constant for concealment probabilities that satisfy the equilibrium condition $\mu_2(0) = \mu_2(\varnothing)$. Since in equilibrium the likelihood ratio after an empty message must be equal to the above, the common posterior after messages $m = \{0, \varnothing\}$ is independent of γ_H. Therefore, by assumption 1 the high-type chooses $\gamma_H^* = 0$ and reveals every review.

Substituting $\gamma_H^* = 0$ into the equilibrium expression for γ_L gives:

$$\gamma_L^*(\lambda_1) = \frac{(1 - \lambda_1)(s_H - s_L)}{(1 - s_L)(1 - \lambda_1 s_H)} \in [0, 1].$$

The manipulation probability $\gamma_L^*(\lambda_1)$ decreases in the probability of trade:

$$\frac{d\gamma_L^*(\lambda_1)}{d\lambda_1} = -\frac{(s_H - s_L)(1 - s_H)}{(1 - s_L)(1 - \lambda_1 s_H)^2} < 0,$$

where the strict inequality follows from: $0 < s_L < s_H < 1$. Then:

$$\gamma_L^*(1) = 0.$$

It remains to consider the boundary case when every bad review is concealed. If $\gamma_H = 1$, the equilibrium relation between concealment probabilities implies $\gamma_L = 1$. A message containing a bad review is then off path, and:

$$\Pr(m = \varnothing \mid s_i) = 1 - \lambda_1 s_i.$$

Consequently, the likelihood ratio following the empty message is given by:

$$\frac{\Pr(m = \varnothing \mid s_H)}{\Pr(m = \varnothing \mid s_L)} = \frac{1 - \lambda_1 s_H}{1 - \lambda_1 s_L}.$$

This is exactly the common likelihood ratio induced by messages $m \in \{0, \varnothing\}$ in the interior disclosure equilibrium characterized above. Moreover, the probability and posterior associated with a positive review are unchanged. Since $m = 0$ and $m = \varnothing$ induce the same continuation payoff, full concealment generates the same distribution of continuation payoffs in the game. Since the high-type is not benefiting from censoring all bad reviews compared to the case of truthful disclosure, by Assumption 1 the disclosure strategy with $\gamma_H = 1$ is not selected.

□

A.6 Continuation beliefs under manipulation

Consider a pooling price that is chosen by the low type with probability $\alpha \in [0, 1]$. Conditional on observing this price, the posterior probability of the high-type is given by:

$$\mu_1(\alpha) = \frac{\mu_0}{\mu_0 + (1 - \mu_0)\alpha}.$$

The corresponding expected quality is:

$$\bar{s}_1(\alpha) = \mu_1(\alpha) s_H + [1 - \mu_1(\alpha)] s_L.$$

Let:

$$\mu := \mu_1(\alpha), \qquad \bar{s} := \bar{s}_1(\alpha).$$

Applying the result of Lemma 3, good reviews are always disclosed, while equilibrium concealment results in $\mu_2(\varnothing) = \mu_2(0)$. To simplify, the public signal can be represented by a positive review $m = 1$ and no positive evidence $m \in \{0, \varnothing\}$, with associated likelihoods conditional on type s_i:

$$\Pr(1 \mid s_i) = \lambda_1 s_i, \qquad \Pr(\{0, \varnothing\} \mid s_i) = 1 - \lambda_1 s_i.$$

Bayes' rule gives:

$$\mu_2^M(1) = \frac{\mu_1 s_H}{\bar{s}_1}, \qquad \mu_2^M(0) = \mu_2^M(\varnothing) = \frac{\mu(1 - \lambda_1 s_H)}{1 - \lambda_1 \bar{s}}.$$

The corresponding posterior expected qualities are

$$\bar{s}_2^M(m) = s_L + \mu_2^M(m)\Delta s, \text{ for } m \in \{1, 0, \varnothing\}.$$

Conditional on the firm being the high-type, its expected second-period expected quality is:

$$\bar{s}^M_{2,H}(\lambda_1, \alpha) = \lambda_1 s_H \bar{s}^M_2(1) + (1 - \lambda_1 s_H)\bar{s}^M_2(\varnothing) = \bar{s} + \frac{\lambda_1 \mu(1-\mu)^2(\Delta s)^3}{\bar{s}(1 - \lambda_1 \bar{s})}.$$

Holding α fixed, learning slope becomes:

$$D_M(\lambda_1, \alpha) := \frac{\partial \bar{s}^M_{2,H}}{\partial \lambda_1} = \frac{\mu(1-\mu)^2(\Delta s)^3}{\bar{s}(1 - \lambda_1 \bar{s})^2} > 0.$$

Similarly, conditional on being the low-type:

$$\bar{s}^M_{2,L}(\lambda_1, \alpha) = \lambda_1 s_L \bar{s}^M_2(1) + (1 - \lambda_1 s_L)\bar{s}^M_2(\varnothing) = \bar{s} - \frac{\lambda_1 \mu^2(1-\mu)(\Delta s)^3}{\bar{s}(1 - \lambda_1 \bar{s})}.$$

A.7 The pooling cutoff under manipulation

Denote:

$$\bar{s} := \bar{s}^{NM}_1 = \mu_0 s_H + (1 - \mu_0) s_L.$$

Under full pooling, the low-type's pooling IC gap is given by:

$$\Phi^M(p_1, 1) = p_1 \lambda_1(p_1) + \kappa \bar{s}^M_{2,L}(\lambda_1(p_1), 1) - 2\kappa s_L.$$

Differentiating it with respect to price p_1 results in:

$$\Phi^M_{p_1}(p_1, 1) = \lambda_1(p_1) + p_1 \lambda_{1,p_1}(p_1) - \kappa \frac{\mu_0^2(1-\mu_0)(\Delta s)^3}{\bar{s}[1 - \lambda_1(p_1)\bar{s}]^2} \lambda_{1,p_1}(p_1).$$

On the price interval $p_1 \in [0, p^{\text{static}}(\bar{s}^{NM}_1)]$, current revenue is weakly increasing:

$$\lambda_1 + p_1 \lambda_{1,p_1} \geq 0.$$

Moreover, as

$$\lambda_{1,p_1} < 0.,$$

and other remaining terms in the last expression are strictly positive, it follows that

$$\Phi^M_{p_1}(p_1, 1) > 0.$$

Then, the IC gap at full pooling price $p^{static}(\bar{s})$ is strictly greater than zero:

$$\Phi^M \left(p^{static}(\bar{s}), 1\right) \geq \kappa \bar{s} + \kappa s_L - 2\kappa s_L = \kappa(\bar{s} - s_L) > 0.$$

Then consider two possibilities. First, if

$$\Phi^M(0, 1) \geq 0,$$

then the cutoff for pooling by the low-type is

$$\widehat{p}^M = 0.$$

Otherwise, if

$$\Phi^M(0, 1) < 0,$$

then by continuity and strict monotonicity of $\Phi^M(p_1, 1)$ there exists a unique cutoff

$$\widehat{p}^M \in (0, p^{\text{static}}(\bar{s})),$$

such that

$$\Phi^M(\widehat{p}^M, 1) = 0.$$

On the relevant price interval:

$$\Phi^M(p_1, 1) \geq 0 \quad \Longleftrightarrow \quad p_1 \geq \widehat{p}^M.$$

For an interior cutoff $\widehat{p}^M$, the indifference condition $\Phi(\widehat{p}^M, 1) = 0$ can be rewritten as:

$$\widehat{p}^M = \frac{\kappa \left[2s_L - \bar{s}^M_{2,L}(\widehat{p}^M)\right]}{1 - G\left(\frac{\widehat{p}^M}{\bar{s}}\right)}.$$

Finally, compare the pooling incentives across the manipulation and no-manipulation regimes. Using the result from Appendix A.2:

$$\bar{s}^M_{2,L}(\lambda_1) - \bar{s}^{NM}_{2,L}(\lambda_1) = \frac{\lambda_1(1-\lambda_1)\mu_0^2(1-\mu_0)(\Delta s)^3}{(1-\bar{s})(1-\lambda_1\bar{s})} \geq 0.$$

As the first-period revenue is identical across regimes at a fixed full-pooling price:

$$\Phi^M(p_1, 1) \geq \Phi^{NM}(p_1) \implies \widehat{p}^M \leq \widehat{p}^{NM}.$$

A.8 Proof of Proposition 2

Proof. Consider the equilibrium price $p_1^{NM} > 0$ that maximizes the no-manipulation objective of the high-type. The no-manipulation optimum is then interior and satisfies FOC:

$$F^{NM}(p_1^{NM}) = 0.$$

Conditional on full pooling, the high-type's marginal payoff under manipulation is

$$F^M(p_1) = R_1'(p_1) + \kappa D_M(\lambda_1(p_1))\lambda_{1,p_1}(p_1).$$

Evaluating the above at the no-manipulation equilibrium price gives

$$\Delta F = F^M(p_1^{NM}) - F^{NM}(p_1^{NM}) = \kappa\lambda_{1,p_1}(p_1^{NM})\left[D_M(\lambda_1^{NM}) - D_{NM}\right],$$

where λ_1^{NM} denotes the trade probability at no-manipulation price p_1^{NM}. Comparing the expressions for $D_M(\lambda_1)$ and D_{NM} derived in Appendices A.6 and A.2, respectively:

$$D^M(\lambda_1) = \frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}_1^{NM}(1-\lambda_1\bar{s}_1^{NM})^2}, \quad \text{and} \quad D^{NM} = \frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}_1^{NM}(1-\bar{s}_1^{NM})},$$

yields:

$$D^M(\lambda_1^{NM}) > D^{NM} \Longleftrightarrow 1 - \lambda_1^{NM}\bar{s}_1^{NM} < \sqrt{1-\bar{s}_1^{NM}} \Longleftrightarrow \lambda_1^{NM} > \frac{1}{1+\sqrt{1-\bar{s}_1^{NM}}}.$$

As the RHS of the inequality above is constant, denote it by:

$$\lambda^* := \frac{1}{1+\sqrt{1-\bar{s}_1^{NM}}}.$$

Since κ is strictly positive and $\lambda_{1,p_1}(p_1^{NM}) < 0$ is strictly negative, the sign of ΔF depends on the comparison of λ_1^{NM} and λ^*;

$$\lambda_1^{NM} \gtrless \lambda^* \quad \Longleftrightarrow \quad \Delta F \lesseqgtr 0.$$

For an ordering of the high-type preferred full-pooling prices to be determined by the sign of ΔF, a sufficient single-crossing regularity condition on the high-type's pooling pricing problem is required. Let

$$r := \frac{p_1}{\bar{s}}, \quad h(r) := \frac{g(r)}{1-G(r)}, \quad \bar{s} := \bar{s}_1^{NM}, \quad \lambda_1 = 1-G(r), \quad \lambda_{1,p_1} = -\frac{g(r)}{\bar{s}}.$$

The high type's marginal payoff under manipulation can be rewritten as:

$$F^M(p_1) = \frac{g(r)}{\bar{s}}\Psi^M(r),$$

where we define:

$$\Psi^M(r) := \bar{s}\left[\frac{1}{h(r)} - r\right] - \kappa D_M(1-G(r)).$$

Since both $g(r)$ and $\bar{s}$ are greater than zero, $F^M(p_1)$ and $\Psi^M(r)$ have the same sign.

Then, it is sufficient for the marginal payoff to satisfy a single-crossing property that under the regularity condition $\Psi^M(r)$ is strictly decreasing on $r \in [0, r^*]$, where r^* corresponds to the static optimal price. Since $F^M(p_1)$ and $\Psi^M(r)$ have the same sign, such monotonicity of $\Psi^M(r)$ guarantees that $F^M(p_1)$ changes the sign at most once. A sufficient regularity condition for this is given by:

$$\frac{2\kappa A g(r)}{[1-\bar{s}(1-G(r))]^3} < \bar{s}\left[1 + \frac{h'(r)}{h(r)^2}\right] \qquad \text{for all } r \in [0, r^*],$$

where

$$A := \mu_0(1-\mu_0)^2(\Delta s)^3.$$

The above condition comes directly from solving $\Psi^{M\prime}(r) < 0$. First, plugging in the expression for the learning slope:

$$D_M(1-G(r)) = \frac{A}{\bar{s}(1-\bar{s}(1-G(r)))^2}.$$

Then, the expression for $\Psi^M(r)$ takes the following form:

$$\Psi^M(r) = \bar{s}\left[\frac{1}{h(r)} - r\right] - \frac{\kappa A}{\bar{s}(1-\bar{s}(1-G(r)))^2}.$$

The derivative with respect to r gives:

$$\Psi^{M\prime}(r) = -\bar{s}\left[1 + \frac{h'(r)}{h(r)^2}\right] + \frac{2\kappa A g(r)}{[1-\bar{s}(1-G(r))]^3}.$$

Then, $\Psi^{M'}(r) < 0$ requires the regularity condition showed above.

From the FOC of the static problem, at static optimal r^*:

$$\frac{1}{h(r^*)} - r^* = 0,$$

which implies that:

$$\Psi^M(r^*) = -\kappa D^M(1 - G(r^*)) < 0.$$

Under the regularity condition, either $\Psi^M(0) \leq 0$, for which the unique full-pooling maximizer is $p_1 = 0$, or $\Psi^M(0) > 0$ and there exists a unique interior root:

$$\Psi^M(r) = 0.$$

In the latter case $F^M(p_1)$ is positive below the root r and negative above it. The high type's full-pooling payoff is then single-peaked and has a unique global maximizer.

Finally, let $p_1^{M,FP}$ denote the high-type's preferred price on the full pooling branch under manipulation. Since

$$F^M(p_1^{NM}) = \Delta F,$$

there are three possible scenarios:

1. $\Delta F < 0 \implies p_1^{M,FP} < p_1^{NM}$,
2. $\Delta F = 0 \implies p_1^{M,FP} = p_1^{NM}$.
3. $\Delta F > 0 \implies p_1^{M,FP} > p_1^{NM}$.

Whenever the high-type's preferred pooling price satisfies:

$$p_1^{M,FP} \geq \widehat{p}^M,$$

the low type's pooling IC is satisfied, and the candidate price is a full-pooling PBE price. Combining the above results yields:

$$\lambda_1^{NM} > \lambda^* \Longrightarrow p_1^{M,FP} < p_1^{NM},$$

and

$$\lambda_1^{NM} < \lambda^* \Longrightarrow p_1^{M,FP} > p_1^{NM}.$$

□

A.9 The cutoff-connected semi-separating branch

For an arbitrary pooling probability α, define the low-type IC gap under manipulation as:

$$\Phi^M(p_1, \alpha) = p_1\lambda_1(p_1, \alpha) + \kappa\bar{s}_{2,L}^M(\lambda_1(p_1, \alpha), \alpha) - 2\kappa s_L.$$

Let:

$$\mu := \mu_1(\alpha), \qquad \bar{s} := \bar{s}_1(\alpha).$$

Using the expression from Appendix A.6:

$$\frac{\partial \bar{s}^M_{2,L}}{\partial \lambda_1} = -\frac{\mu^2(1-\mu)(\Delta s)^3}{\bar{s}(1-\lambda_1 \bar{s})^2} < 0,$$

the derivative with respect to initial price p_1 at fixed α gives:

$$\Phi^M_{p_1}(p_1, \alpha) = \lambda_1 + p_1 \lambda_{1,p_1} - \kappa \frac{\mu^2(1-\mu)(\Delta s)^3}{\bar{s}(1-\lambda_1 \bar{s})^2} \lambda_{1,p_1}.$$

The expected quality of the firm in $t = 1$ at fixed α is given by:

$$\bar{s}_{1,\alpha} = -\frac{\mu_0(1-\mu_0)(s_H - s_L)}{[\mu_0 + (1-\mu_0)\alpha]^2} < 0.$$

For any pooling probability such that $\alpha < 1$ the following is true: $\alpha \leq 1$,

$$\bar{s}_1(\alpha) > \bar{s}_1(1),$$

which implies:

$$p^{static}(\bar{s}_1(\alpha)) > p^{static}(\bar{s}_1(1)).$$

For some $\alpha < 1$, the semi-separating price p_1 satisfies:

$$p_1 \leq \widehat{p}^M < p^{static}(\bar{s}_1(1)) \leq p^{static}(\bar{s}_1(\alpha)).$$

Since the price p_1 lies below the one-shot game optimal price, the current revenue is increasing at fixed α:

$$\lambda_1 + p_1 \lambda_{1,p_1} > 0,$$

where: $\lambda_{1,p_1} < 0$. It follows that the IC gap is increasing in p_1:

$$\Phi^M_{p_1}(p_1, \alpha) > 0.$$

By construction, at the full-pooling cutoff the IC gap is given by:

$$\Phi^M(\widehat{p}^M, 1) = 0.$$

Then, because the IC gap is also strictly increasing in price p_1 at the cutoff, $\Phi^M_{p_1}(\widehat{p}^M, 1) > 0$, the implicit-function theorem locally defines a differentiable function:

$$p_1 = p^M(\alpha), \qquad p^M(1) = \widehat{p}^M.$$

Using the IFT gives:

$$\frac{dp^M(\alpha)}{d\alpha} = -\frac{\Phi^M_\alpha(p^M(\alpha), \alpha)}{\Phi^M_{p_1}(p^M(\alpha), \alpha)}.$$

For the cutoff price $p^M(\alpha) = \widehat{p}^M$ to be the lowest price for which full-pooling occurs, it requires $\alpha < 1$ for some price marginally below $\widehat{p}^M$. I.e. the LHS needs to be greater than zero. As $\Phi^M_{p_1}(p^M(\alpha), \alpha) > 0$, this requires that locally:

$$\Phi^M_\alpha(\widehat{p}^M, 1) < 0.$$

As G is continuously differentiable and the continuation beliefs are continuous on the pooling branch, $\Phi^M(p_1, \alpha)$ is continuously differentiable in a neighborhood of $(\widehat{p}^M, 1)$. Then by continuity, in a one-sided neighborhood of the cutoff the following holds:

$$\frac{dp^M(\alpha)}{d\alpha} > 0.$$

Thus, for α sufficiently close to one:

$$p^M(\alpha) < \widehat{p}^M.$$

Equivalently, it can be rewritten locally as:

$$\alpha = \alpha^M(p_1),$$

where:

$$\alpha^{M'}(p_1) = -\frac{\Phi^M_{p_1}(p_1, \alpha^M(p_1))}{\Phi^M_{\alpha}(p_1, \alpha^M(p_1))} > 0.$$

Therefore, locally along the cutoff-connected branch, a decrease in price p_1 leads to a decrease in a pooling probability $\alpha^M(p_1)$.

A.10 Joint non-emptiness of the maintained parameter regions

Under Assumption 2 and the regularity condition imposed, the set of admissible primitives is constrained. This section shows that the intersection of the admissible primitives with the set of parameters for which PBE described in Proposition 3 exist is non-empty, therefore for some primitives the equilibrium behavior described occurs.

Let Ω denote the set of admissible primitives, and define:

$$\Omega^{FP} := \left\{\omega \in \Omega : \Phi^{NM}\left(p_1^{NM,FP}(\omega)\right) \geq 0\right\}.$$

as the set of primitives satisfying the full-pooling no-manipulation benchmark in Assumption 2. Then, let denote the set of primitives satisfying the full-pooling regularity condition as:

$$\Omega^{Reg} := \left\{\omega \in \Omega : \Psi^M(r) \text{ is strictly decreasing on } [0, r^*]\right\},$$

And finally, let denote the set of primitives for which the low-type's IC gap is decreasing in α at the pooling cutoff:

$$\Omega^{SS} := \left\{\omega \in \Omega : \Phi^M_{\alpha}(\widehat{p}^M, 1) < 0\right\}.$$

Suppose the distribution of consumer's valuation is uniform $G(v) = v$. Denote the difference between the types as $\Delta s := s_H - s_L$, and and consider prior probability such that $\mu_0 \to 0$. For the uniform distribution:

$$r^* = \frac{1}{2}, \qquad \kappa = \frac{1}{4}.$$

First step is to verify the non-emptiness arising from the Assumption 2. Under no-manipulation the full pooling is assumed and results in $\bar{s} = s_L + \mu_0 \Delta s$. Given the uniform distribution $G(v)$, the high-type's unconstrained full-pooling price is:

$$p_1^{NM} = \frac{\bar{s} - \kappa D_{NM}}{2},$$

where the no-manipulation learning slope takes the usual form: $D_{NM} = \frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}(1-\bar{s})}$. For every sufficiently small $\mu_0 > 0$, this price is strictly positive and interior. The low-type's no-manipulation reputation coefficient is given by:

$$A_L = \frac{\mu_0^2(1-\mu_0)(\Delta s)^3}{\bar{s}(1-\bar{s})}.$$

Substituting the above p_1^{NM} into the low-type's pooling IC gap gives:

$$\Phi^{NM}(p_1^{NM}) = 2\kappa(\bar{s} - s_L) - \frac{\kappa^2 (D_{NM})^2}{4\bar{s}} - \kappa A_L \lambda_1^{NM}.$$

Using the definition of $\bar{s}$, the first bracket in the RHS can be replace with $\mu_0 \Delta s$:

$$\frac{\Phi^{NM}(p_1^{NM})}{\mu_0} = 2\kappa\Delta s - \frac{\kappa^2}{4\bar{s}} \frac{(D^{NM})^2}{\mu_0} - \kappa \frac{A_L}{\mu_0} \lambda_1^{NM}.$$

For μ_0 going to zero, the above can be simplified as follows. Notice that:

$$\lim_{\mu_0 \to 0} \frac{(D^{NM})^2}{\mu_0} = \frac{\mu_0(1-\mu_0)^4(\Delta s)^6}{\bar{s}^2(1-\bar{s})^2} \longrightarrow 0,$$

and as $\kappa \in [0,1]$ and $\lambda_1^{NM} \in [0,1]$:

$$\lim_{\mu_0 \to 0} \kappa \frac{A_L}{\mu_0} \lambda_1^{NM} = \kappa \frac{\mu_0(1-\mu_0)(\Delta s)^3}{\bar{s}(1-\bar{s})} \lambda_1^{NM} \longrightarrow 0.$$

Therefore, the last components result in:

$$\lim_{\mu_0 \to 0} \frac{\Phi^{NM}(p_1^{NM})}{\mu_0} = 2\kappa\Delta s > 0,$$

and this implies $\Phi^{NM}(p_1^{NM}) > 0$, for sufficiently small $\mu_0 > 0$. Hence, Assumption 2 holds and the set of primitives for which the Assumption 2 holds is non-empty, $\Omega^{FP} \neq \emptyset$.

Next verify the full-pooling regularity condition. For the uniform distribution, from the Appendix A.8 it is sufficient that:

$$\kappa A < \bar{s}(1-\bar{s})^3,$$

where

$$A = \mu_0(1-\mu_0)^2(\Delta s)^3.$$

As $\mu_0 \to 0$, the LHS of the above inequality goes to zero

$$\kappa A \longrightarrow 0,$$

while the RHS is strictly above zero in the limit as $0 < s_L < 1$:

$$\bar{s}(1-\bar{s})^3 \longrightarrow s_L(1-s_L)^3 > 0.$$

Hence the full-pooling regularity condition also holds for a prior belief sufficiently close to zero. The admissible set of primitives for which the regularity condition is satisfied is therefore non-empty, $\Omega^{Reg} \neq \emptyset$.

Then, it is necessary to establish whether the set of primitives for which the cutoff-connected semi-separating equilibrium exists. Let denote the belief conditional on observed pooling branch price, at pooling probability α, as $\mu_1(\alpha) := \mu$. Rewrite the low-type's pooling IC gap as a function of that belief:

$$\bar{s}(\mu) = s_L + \mu\Delta s, \qquad \lambda(p_1, \mu) = 1 - \frac{p_1}{\bar{s}(\mu)},$$

and using the expression from Appendix A.6:

$$\Phi^M(p_1, \mu) = p_1\lambda(p_1, \mu) + \kappa\left[\bar{s}(\mu) - \frac{\lambda(p_1, \mu)\mu^2(1-\mu)(\Delta s)^3}{\bar{s}(\mu)[1 - \lambda(p_1, \mu)\bar{s}(\mu)]}\right] - 2\kappa s_L.$$

At full pooling, $\mu = \mu_0$, and for $\mu_0 \to 0$, as the second expression in the bracket goes to zero the above converges on the relevant price interval to:

$$p_1\left(1 - \frac{p_1}{s_L}\right) - \kappa s_L.$$

As under the uniform distribution $\kappa = \frac{1}{4}$, at the limit the price that guarantees $\Phi^M(p_1, \mu) = 0$ is $p_1 = \frac{s_L}{2}$. It follows that the full-pooling manipulation cutoff satisfies:

$$\widehat{p}^M \longrightarrow \frac{s_L}{2}.$$

Differentiating the expression for $\Phi^M(p_1, \mu)$ with respect to μ, holding p_1 fixed, gives:

$$\lim_{\mu_0 \to 0} \frac{\partial \Phi^M(p_1, \mu)}{\partial \mu}\bigg|_{p_1 = \widehat{p}^M,\, \mu = \mu_0} = \frac{\Delta s}{4} + \kappa\Delta s = \frac{\Delta s}{2}.$$

The first term is the limiting effect through current revenue, while the second is the limiting effect through continuation value. The remaining manipulation term has a factor μ^2, and its derivative converges to zero. Pooling becomes more profitable for the low-type as the belief increases.

At $\alpha = 1$, $\mu_{1,\alpha} = -\mu_0(1 - \mu_0)$. Then the derivative of an IC gap divided by prior gives:

$$\frac{\Phi^M_\alpha(\widehat{p}^M, 1)}{\mu_0} = \frac{\mu_{1,\alpha}}{\mu_0}\frac{\partial \Phi^M(p_1, \mu)}{\partial \mu} = -(1 - \mu_0)\left.\frac{\partial \Phi^M(p_1, \mu)}{\partial \mu}\right|_{p_1 = \widehat{p}^M,\, \mu = \mu_0}.$$

Then, checking the behavior at the limit results in:

$$\lim_{\mu_0 \to 0} \frac{\Phi^M_\alpha(\widehat{p}^M, 1)}{\mu_0} = -\frac{\Delta s}{2} < 0.$$

Since μ_0 is positive, the above implies:

$$\Phi^M_\alpha(\widehat{p}^M, 1) < 0$$

for a μ_0 sufficiently close to zero. By Appendix A.9, there exists local continuation into the semi-separating region for some set of primitives, $\Omega^{SS} \neq \emptyset$.

Once the no-emptiness of the three sets is established, it is important to determine whether

there is some non-empty set of primitives in the intersection of the these, $\Omega^{FP} \cap \Omega^{Reg} \cap \Omega^{SS}$. The three preceding arguments use the same family of primitives: $G(v) = v$, and a sufficiently small positive μ_0, and the three conditions can be satisfied simultaneously. In particular, there exists $\bar{\mu}_0 > 0$ such that, for every $\mu_0 \in (0, \bar{\mu}_0)$, the corresponding primitive vector belongs to the intersection, and therefore:

$$\Omega^{FP} \cap \Omega^{Reg} \cap \Omega^{SS} \neq \emptyset.$$

Then, it is necessary to establish whether for the set of admissible primitives all of the defined equilibria forms can be sustained. For primitives in the intersection $\Omega^{FP} \cap \Omega^{Reg}$, Assumption 2 states that the no-manipulation benchmark takes the form of full-pooling. Under manipulation, Appendix A.7 establishes a unique pooling cutoff $\widehat{p}^M$, and the low-type pools for every price weakly above the cutoff.

The high type's full-pooling payoff is continuous on the compact admissible price interval, and the regularity condition implies a unique high-type preferred price conditional on pooling. Consequently, a full-pooling PBE exists for the admissible set of primitives.

As it was already showed that the intersection $\Omega^{FP} \cap \Omega^{Reg} \cap \Omega^{SS}$ is non-empty, Appendix A.9 implies that the low type's indifference condition locally defines a cutoff-connected semi-separating branch satisfying:

$$p_1 < \widehat{p}^M, \qquad \alpha^M \in (0, 1).$$

Some semi-separating solutions exist locally below the full-pooling cutoff. Any semi-separating solution satisfies $\Phi^M(p_1, \alpha^M) = 0$, which implies that the low type obtains payoff equal to:

$$U_L^{SS} = 2\kappa s_L.$$

From Appendix A.6, the expected posterior belief for the high-type is strictly higher on the pooling branch:

$$\bar{s}_{2,H}^M - \bar{s}_{2,L}^M = \frac{\lambda_1 \mu (1-\mu)(\Delta s)^3}{\bar{s}(1 - \lambda_1 \bar{s})} > 0.$$

Since on that branch the initial revenue is the same for both, the overall payoff is greater for the high-type:

$$U_H^{SS} > U_L^{SS} = 2\kappa s_L.$$

Under the maintained off-path beliefs, any deviation from the pooling branch price gives the high-type payoff of at most $2\kappa s_L$. No profitable deviation combined with the disclosure strategy characterized in Lemma 3 support a semi-separating PBE.

The same non-empty set of primitives satisfies Assumption 2, the full-pooling regularity condition, and the local cutoff-condition, and support both full-pooling and semi-separating PBE.

A.11 Proof of Lemma 4

Proof. Suppose that the two types choose distinct first-period prices – some separating PBE – and obtain corresponding equilibrium payoffs given by V_H^S and V_L^S. In separating equilibrium each price determines unambiguously the type and the posterior over the type is degenerated. Since any review is on-path for each type, by mimicking other type's action a firm can receive exactly the payoff associated with the mimicked price path.

To sustain a separating PBE incentive compatibility for both types requires:

$$V_H^S \geq V_L^S \qquad \text{and} \qquad V_L^S \geq V_H^S.$$

The above conditions can both be satisfied only for:

$$V_H^S = V_L^S.$$

The best revenue V_L^S a low-type firm can secure is equal to double the optimal static revenue at expected quality s_L. Therefore:

$$V_L^S \leq 2\kappa s_L.$$

At the assumed off-path beliefs, the low-type can choose to deviate from any price to $p^{static}(s_L)$ and guarantee itself exactly:

$$2\kappa s_L.$$

Then, every fully separating PBE needs to satisfy:

$$V_H^S = V_L^S = 2\kappa s_L.$$

Then, consider a full pooling PBE at the initial price:

$$p^{static}(\bar{s}_1).$$

Both types obtain first-period revenue $\kappa\bar{s}_1$, and continuation revenue of at least κs_L. Therefore, the full pooling PBE at that price guarantees both types payoff of strictly more than $2\kappa s_L$., as:

$$\kappa(\bar{s}_1 + s_L) > 2\kappa s_L.$$

At the pessimistic off-path beliefs such pooling profile constitutes a PBE. It then strictly payoff-dominates every fully separating PBE for both types. Hence, if the fully separating equilibrium exists, it is never selected. □

A.12 Proof of Proposition 3

Proof. First, lets consider $p_1^{NM} = 0$, for which $\lambda_1^{NM} = 1$. Appendices A.2 and A.6 imply:

$$D^M(1) > D^{NM}.$$

Since $\lambda_{1,p_1}(0) < 0$ and $F^{NM}(0) \leq 0$, it follows that:

$$F^M(0) < F^{NM}(0) \leq 0.$$

Under the full-pooling regularity condition, the high type's full-pooling payoff is single-peaked. The unique high-type preferred full-pooling price under manipulation is:

$$p_1^M = 0.$$

Moreover, Appendix A.7 and Assumption 2 imply:

$$\widehat{p}^{NM} = \widehat{p}^M = 0.$$

Then, suppose $p_1^{NM} > 0$. Appendix A.7 establishes that full pooling is feasible on the interval:

$$[\widehat{p}^M, p^{static}(\bar{s}_1^{NM})].$$

Let p_1^M denote the unique high-type preferred price conditional on full pooling characterized in Appendix A.8. If the unconstrained full-pooling maximizer lies below or at $\widehat{p}^M$, the single-peakedness of the objective function implies the unique maximizer is $p_1^M = \widehat{p}^M$. Otherwise, by Proposition 2:

1. If $\lambda_1^{NM} < \lambda^*$, then:
$$p_1^M > p_1^{NM} \geq \widehat{p}^{NM} \geq \widehat{p}^M.$$

2. If $\lambda_1^{NM} = \lambda^*$, then:
$$p_1^M = p_1^{NM}.$$

3. If $\lambda_1^{NM} > \lambda^*$, then:
$$p_1^M < p_1^{NM}.$$

   By construction $p_1^M \geq \widehat{p}^M$. As $p_1^{NM} > 0$, the Appendix A.7 gives:
$$\Phi^M(p_1^{NM}, 1) > \Phi^{NM}(p_1^{NM}) \geq 0.$$

   By strict monotonicity of IC gap at full pooling, $\widehat{p}^M < p_1^{NM}$. Then: $\widehat{p}^M \leq p_1^M < p_1^{NM}$.

Appendix A.9 characterizes the local cutoff-connected semi-separating branch. Any interior high-type preferred pair (p_1^M, α^M) satisfies:

1. $\Phi^M(p_1^M, \alpha^M) = 0$
2. $F^M(p_1^M, \alpha^M) + B_1(p_1^M) + B_2(p_1^M) = 0.$

The first condition guarantees that the low-type is indifferent between pooling and separating, therefore may choose either with positive probability in a PBE. The high-type preferred price on the semi-separating equilibrium branch satifies the second condition. Moreover, for such p_1^M the following holds:
$$p_1^M < \widehat{p}^M \leq \widehat{p}^{NM} \leq p_1^{NM}.$$
Appendix A.10 establishes that under the set of primitives satisfying Assumption 2 and the regularity condition is jointly non-empty and that both full-pooling and semi-separating PBE exist for some admissible primitives.

Finally, Lemma 4 shows that every fully separating PBE is strictly payoff-dominated for both types and therefore cannot be selected.

Among the remaining full-pooling and semi-separating PBE, the selection criterion chooses an equilibrium that maximizes the high-type's payoff. □

A.13 Proof of Proposition 4

Proof. Consider the following expression:
$$\Delta F = \kappa \lambda_{1,p_1}(p_1^{NM}) \left[D^M(\lambda_1^{NM}) - D^{NM} \right],$$
and use the fact that:
$$\lambda_{1,p_1}(p_1^{NM}) = -\frac{g(r^{NM})}{\bar{s}_1}.$$
Then, take the expressions for learning slopes under manipulation and no-manipulation, respectively:
$$D^M(\lambda_1^{NM}) = \frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}_1\left(1-\lambda_1^{NM}\bar{s}_1\right)^2}, \quad \text{and} \quad D^{NM} = \frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}_1(1-\bar{s}_1)},$$

and subsitute them into the formula for ΔF, and take the common part out of the last bracket:

$$\Delta F = -\frac{\kappa g(r^{NM})}{\bar{s}_1}\left[\frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}_1}\right]\left[\frac{1}{(1-\lambda_1^{NM}\bar{s}_1)^2}-\frac{1}{1-\bar{s}_1}\right].$$

Note that for $\lambda_1^{NM} = \lambda^*$ the last bracket is equal to 0. Consider $\lambda_1^{NM} \neq \lambda^*$. Substitute the expression for λ_1^{NM} using $\lambda_1^{NM} = 1 - G(r^{NM})$:

$$\Delta F = -\frac{\kappa g(r^{NM})}{\bar{s}_1}\left[\frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}_1}\right]\left[\frac{1}{(1-(1-G(r^{NM}))\bar{s}_1)^2}-\frac{1}{1-\bar{s}_1}\right]. \tag{2}$$

Holding all remaining terms in equation (2) fixed, define:

$$C := \frac{\kappa\mu_0(1-\mu_0)^2}{\bar{s}_1^2}\left|\frac{1}{(1-(1-G(r^{NM}))\bar{s}_1)^2}-\frac{1}{1-\bar{s}_1}\right| > 0,$$

where the last component is kept in absolute values to isolate the effect on size of ΔF from the sign effect. Then:

$$|\Delta F| = C\, g(r^{NM})(\Delta s)^3.$$

From the above definition is follows that:

$$\lim_{\Delta s \to 0} |\Delta F| = 0.$$

Then, taking derivatives of ΔF with respect to local density of demand and difference in quality of types results in:

$$\frac{\partial|\Delta F|}{\partial g(r^{NM})} = C(\Delta s)^3 = \frac{|\Delta F|}{g(r^{NM})} > 0,$$

and:

$$\frac{\partial|\Delta F|}{\partial \Delta s} = 3Cg(r^{NM})(\Delta s)^2 = \frac{3|\Delta F|}{\Delta s} > 0.$$

The above shows the effects described in part 1. of Proposition 4.

Then, consider the effect of change in λ_1^{NM} on ΔF holding all other elements fixed:

$$\frac{\partial \Delta F}{\partial \lambda_1^{NM}} = -\frac{2\kappa g(r^{NM})}{(1-\lambda_1^{NM}\bar{s}_1)^3}\left[\frac{\mu_0(1-\mu_0)^2(\Delta s)^3}{\bar{s}_1}\right] < 0.$$

Higher benchmark trade level increases the downward local price force.

Next, consider the threshold determining the direction of the price response under manipulation:

$$\lambda^* = \frac{1}{1+\sqrt{1-\bar{s}_1}}.$$

Differentiating with respect to expected quality gives:

$$\frac{\partial \lambda^*}{\partial \bar{s}_1} = \frac{1}{2\sqrt{1-\bar{s}_1}\left(1+\sqrt{1-\bar{s}_1}\right)^2} > 0.$$

Holding everything else fixed, a higher expected quality $\bar{s}_1$ raises the threshold required for manipulation to generate a downward price force. The above shows the effects described in part 2. of Proposition 4.

□

A.14 Proof of Lemma 5

Proof. First, start by excluding possibility of fabricating bad reviews by either type. The likelihood of generating an honest bad review is greater for the low-type, therefore absent of manipulation the following holds $\mu_2(1) > \mu_2(\varnothing) > \mu_2(0)$. Suppose the low-type fakes bad reviews with positive probability. Since this shifts more probability weight on the message $m = 0$, this lowers the $\mu_2(0)$ farther. Then the resulting posterior belief leads to a lower posterior belief, and is therefore payoff-dominated by no manipulation.

Then suppose the high-type fabricates a bad review with positive probability. For such strategy to be a part of PBE, it requires the resulting posterior beliefs to satisfy:

$$\mu_2(\varnothing) = \mu_2(0),$$

otherwise the high-type would have a profitable deviation to faking more or less. The low-type best response is to generate fake good reviews since:

$$\mu_2(1) > \mu_2(\varnothing) = \mu_2(0).$$

But for the associated faking strategies to be PBE, the probability ρ_L has to yield $\mu_2(1) = \mu_2(\varnothing)$, otherwise the low-type would either prefer to fake less (for $\mu_2(1) < \mu_2(\varnothing)$) or more (for $\mu_2(1) > \mu_2(\varnothing)$). To sustain a PBE with a positive probability of generating bad reviews, it requires the resulting posteriors to satisfy:

$$\mu_2(1) = \mu_2(\varnothing) = \mu_2(0),$$

and such babbling faking strategy is payoff-dominated for the high-type by a truthful strategy ρ_H^0.

After ruling out strategies assigning positive probability to faking bad reviews, denote by

$$\rho_i := \rho_{i,1}, \qquad i \in \{L, H\},$$

the probability that type i generates a fake positive review conditional on $y = \varnothing$.

Conditional on type s_i, the likelihoods of messages $m = 1$ and $m = \varnothing$ are:

$$\Pr(m = 1 \mid s_i) = \lambda_1 s_i + (1 - \lambda_1)\rho_i, \quad \text{and} \quad \Pr(m = \varnothing \mid s_i) = (1 - \lambda_1)(1 - \rho_i).$$

Denote the belief after observing a pooling price as: $\mu_1 \in (0, 1)$. Using Bayes' rule gives:

$$\mu_2(1) = \frac{\mu_1 \left[\lambda_1 s_H + (1 - \lambda_1)\rho_H\right]}{\mu_1 \left[\lambda_1 s_H + (1 - \lambda_1)\rho_H\right] + (1 - \mu_1)\left[\lambda_1 s_L + (1 - \lambda_1)\rho_L\right]},$$

and

$$\mu_2(\varnothing) = \frac{\mu_1(1 - \lambda_1)(1 - \rho_H)}{\mu_1(1 - \lambda_1)(1 - \rho_H) + (1 - \mu_1)(1 - \lambda_1)(1 - \rho_L)}.$$

In any PBE under manipulation, the equilibrium condition requires that:

$$\mu_2(\varnothing) = \mu_2(1).$$

Otherwise, if $\mu_2(\varnothing) > \mu_2(1)$ the low type would profitably deviate to fabricating with lower probability, and conversely, if $\mu_2(\varnothing) < \mu_2(1)$ the low type would prefer to deviate to more faking. Equating and simplifying the expression for $\mu_2(1)$ and $\mu_2(\varnothing)$ gives:

$$\lambda_1(s_H - s_L) + (1 - \lambda_1 + \lambda_1 s_L)\rho_H - (1 - \lambda_1 + \lambda_1 s_H)\rho_L = 0.$$

This can be rewritten for ρ_L as a function of λ_1 and ρ_H:

$$\rho_L(\lambda_1, \rho_H) = \frac{\lambda_1(s_H - s_L) + (1 - \lambda_1 + \lambda_1 s_L)\rho_H}{1 - \lambda_1 + \lambda_1 s_H}.$$

As ρ_H increases so does the probability of faking by the low-type. Any change in faking probability by high-type is offset, and the posteriors on-path $\mu_2(\varnothing) = \mu_2(1)$. Since for the high-type the combined probability of generating message $m = 1$ or $m = \varnothing$ is independent of ρ_H:

$$Pr(m = 1 \vee m = \varnothing | s_H) = \lambda_1 s_H + (1 - \lambda_1)\rho_H + (1 - \lambda_1)(1 - \rho_H) = 1 - \lambda_1(1 - s_H),$$

then by the truth-leaning assumption the high-type sets $\rho_H = 0$. The low-type's faking good review probability is:

$$\rho_L^*(\lambda_1) = \frac{\lambda_1(s_H - s_L)}{1 - \lambda_1 + \lambda_1 s_H}.$$

Moreover, the low type's probability of fabricating a good review ρ_L is strictly increasing in first-period trade:

$$\frac{d\rho_L^*(\lambda_1)}{d\lambda_1} = \frac{s_H - s_L}{(1 - \lambda_1 + \lambda_1 s_H)^2} > 0.$$

□

A.15 Proof of Proposition 6

Proof. Define:

$$\mu := \mu_1, \quad \bar{s} := \bar{s}_1, \quad A := \mu(1 - \mu)^2(\Delta s)^3.$$

By Lemma 5, under fake reviews regime the messages $m \in \{1, \varnothing\}$ induce the same posterior belief. Using Bayes' rule, the high type's expected posterior quality is given by:

$$\bar{s}_{2,H}^F(\lambda_1) = \bar{s} + \frac{\lambda_1 A}{(1 - \bar{s})[1 - \lambda_1(1 - \bar{s})]}.$$

Therefore, the learning slope under fake reviews regime is:

$$D_F(\lambda_1) = \frac{A}{(1 - \bar{s})[1 - \lambda_1(1 - \bar{s})]^2}.$$

Comparing this expression with no-manipulation benchmark slope $D^{NM} = \frac{A}{\bar{s}(1-\bar{s})}$, gives:

$$D_F(\lambda_1) \gtrless D^{NM} \quad \iff \quad \lambda_1 \gtrless \lambda_F^* := \frac{1}{1 + \sqrt{\bar{s}}}.$$

The threshold for censoring regime is established in Proposition 2, and denoted as:

$$\lambda_C^* = \frac{1}{1 + \sqrt{1 - \bar{s}}}.$$

At the expected quality $\bar{s} > \frac{1}{2}$, the resulting thresholds across regimes satisfy:

$$\lambda_F^* < \lambda_C^*.$$

The maintained regularity condition required $\Psi^{M'}(r) < 0$. The derivative of the corresponding single-crossing function is given by:

$$\Psi^{k'}(r) = -\bar{s}\left[1 + \frac{h'(r)}{h(r)^2}\right] + \kappa g(r) D'_k(\lambda_1),$$

where the first expression is negative and the second is positive. Comparison of derivatives of learning slopes under regimes F and C, respectively, at $\bar{s}_1 > \frac{1}{2}$ gives:

$$D'_F(\lambda_1) = \frac{2A}{[1 - \lambda_1(1 - \bar{s})]^3} \leq \frac{2A}{(1 - \lambda_1 \bar{s})^3} = D'_C(\lambda_1),$$

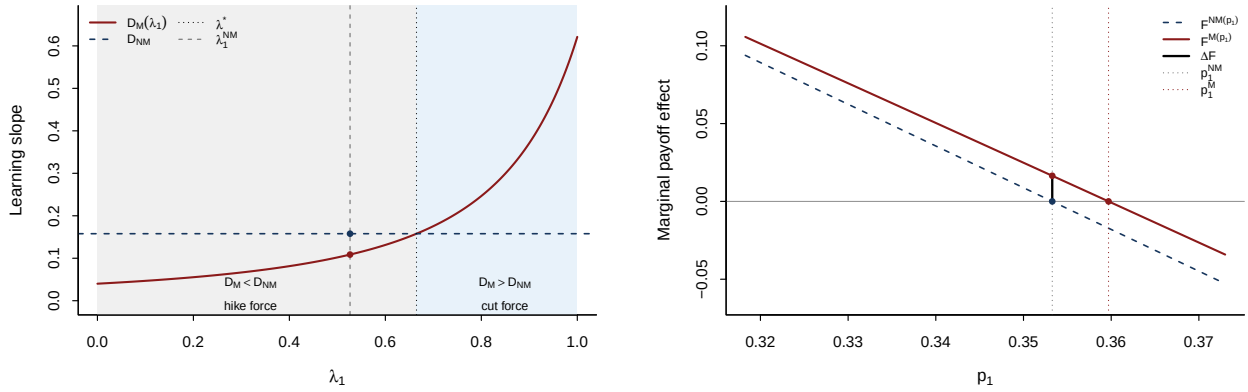
As the value of the derivative for the learning slope under fake review regime is lower, therefore the positive expression is smaller compared to censoring. The regularity condition is therefore satisfied.

This implies that the full-pooling pricing argument from Proposition 2 can be applied to compare the effect of both regimes on the price change direction:

$$\lambda_1^{NM} \gtrless \lambda_k^* \quad \Longleftrightarrow \quad p_1^k \lesseqgtr p_1^{NM}, \qquad k \in \{C, F\}.$$

Combining the above results gives the three cases stated in the proposition. The welfare implications follow from Proposition 5. □

B Additional Numerical Illustration



(a) Learning slopes

(b) Shift in the pricing first-order condition

Figure 3: Pooling price response to review censoring at: $G = U[0, 1]$, $\mu_0 = 0.7$, $s_L = 0.2$, $s_H = 0.98$, corresponding to the price hike case. Panel (a) compares learning slopes $D_M(\lambda_1)$ and D_{NM}. The intersection at λ^* separates the price-increase and price-reduction regions. Panel (b) illustrates $\Delta F > 0$ resulting in a upward shift of the pooling price.